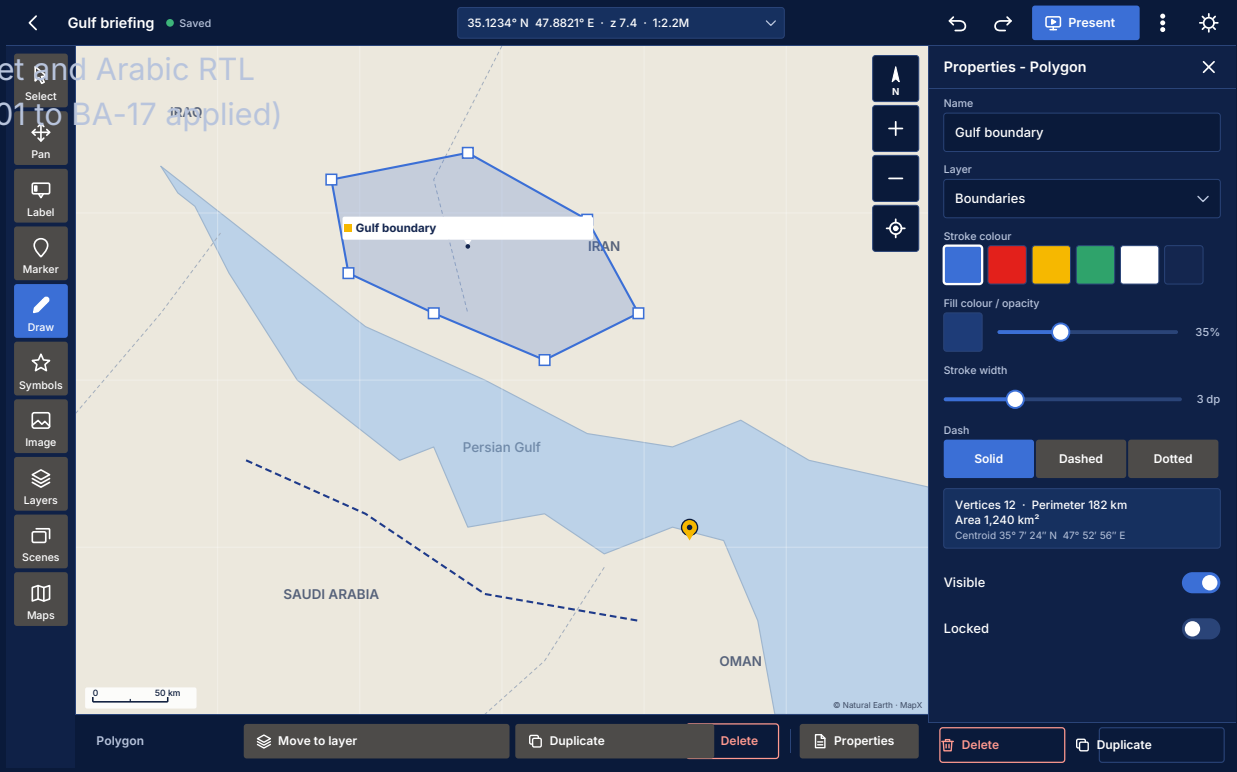
MapX

UI/UX Design Guide

Android implementation reference for phone, tablet and Arabic RTL
Dark and light themes · Layout Blueprint v1.1 (BA-01 to BA-17 applied)



Version 1.1 · 5 November 2026 · Native Android · Kotlin + Jetpack Compose · LTR + RTL

About this guide

The implementation reference for the MapX Android interface: every screen, region, component, state and dp value an Android team needs, derived from the Master Project Specification v1.0 and the Layout Blueprint v1.

How to read it

- Sections 1–15 follow the order of the design brief. Appendix A lists the 17 Blueprint amendments with where each one lives; Appendix B is the validation record.
- All sizes are dp unless stated; text sizes are sp. Mockups are vector and zoomable; red badges are annotations, not UI.
- Mockups show English LTR unless a caption says RTL. Arabic strings are illustrative translations for layout testing, not final copy.
- The mockups draw each theme faithfully but use Inter and DejaVu Sans as stand-ins: the Cairo font specified for the app was not available in the authoring environment. Spacing, weights and sizes are specified for Cairo (Section 5).

Source priority applied

- 1. Master Project Specification v1.0 (behaviour, tokens, formats, security).
- 2. Layout Blueprint v1 as amended by Spec §12.4 (structure, regions, dp). This guide is Blueprint v1.1.
- 3. Owner decisions in the design brief (themes, icon style, animation, five-slot dock, 320 dp panel, end-edge controls).

Not built or tested

This is a design document. Nothing here has been implemented or run on a device; contrast ratios were computed, everything else is specification.

Deviations and reconciliations (owner approval requested)

ID	Item	Why
DEV-1	Phone portrait: top bar 56 → 48 dp	Brief problem C. Spec §12.3 lists 56; gives +8 dp of map.
DEV-2	Phone portrait dock 64 → 56 dp; slots Select · Pan · Draw · Layers · More	Owner decision; Marker, Label, Symbols move to More.
DEV-3	Phone landscape: top bar 40 → 48, context bar 44 → 48	FR-ACC-01 and S1-07 require ≥ 48 dp targets in both directions.
DEV-4	Phone landscape rail: Select, Pan, Draw, Layers, Scenes + More	Keeps Pan first-class (owner); Symbols joins More.
DEV-5	Settings gains an Appearance group (theme, animation)	Owner decisions; FR-SET-02 does not list it.
DEV-6	Alert red is the spec token #E3201B	Brief quotes #E83218; spec wins. Both fail as text on navy.
DEV-7	Blueprint Fig. 15 note ("controls to the start edge") is corrected: controls stay at the logical end edge	Law L10; end = left in RTL.
DEV-8	Coordinate popup also hosts Go to coordinates	FR-MAP-05 had no UI; avoids another toolbar item.
DEV-9	"Delete last-but-one" becomes "Delete is pinned directly before the final slot"	Resolves the context-bar conflict (§10).

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Blueprint amendments and resolved problems

The Layout Blueprint v1 is replaced by this guide. Each amendment from Spec §12.4 and each problem from the design review has a concrete resolution.

ID	Amendment (Spec §12.4)	How it is implemented here	Where
BA-01	Rail items 48 dp high	Tablet tiles stay 56 dp; phone-landscape rail items are 48 dp; rail scrolls.	§14, §7
BA-02	S0 Lock/Login + account dialogs	Centred card (tablet) / full-width (phone), no rail/dock/panels; first-run, change password, recovery code, erase-all.	§13.1
BA-03	Remote: no Undo/Redo/Tool; add Start/Exit, Reset view; "for this presentation" toggles	Single scrolling column; landscape two-column; disabled-not-hidden.	§13.7
BA-04	"comments" → "editor-only UI state"	Applied to external-display rules and the presenter/audience figure notes.	§13.6
BA-05	More: Project group; tappable readout → coordinate-format popup	Project menu (Export..., Import..., Backup..., Project properties, Lock now); DD/DDM/DMS/UTM/MGRS popup.	§7, §9
BA-06	Locate control states and stack position	idle, searching, located, follow, denied/off + no-GNSS; stack: compass, +, −, locate.	§9
BA-07	Layers: row overflow + header overflow	Row: Rename, Export layer, Duplicate. Header: Import layer.... Footer fixed 48 dp.	§11
BA-08	Maps panel: format chooser, conversion rows, details actions	Rename, Delete, Export source, Georeference; plain-language hints.	§11
BA-09	Georeference screen	Two synchronised views + point table, RMSE, accept/adjust/reject; tablet and phone.	§13.4
BA-10	Export wizard, import report, passphrase, relink	One ModalLayer dialog, steps inside; progress and report; snackbar with Share.	§13.5
BA-11	Home: account chip with Lock; Import project package...	Chip in header; overflow on Open Project.	§8
BA-12	Settings sections	Phone single column; tablet list + detail; all FR-SET-02 groups.	§13.2
BA-13	Show my location toggle (default off)	Scenes panel → Presentation options.	§9, §11
BA-14	Remote control on/off + pairing QR entry	Scenes panel → Presentation options; opens the pairing dialog.	§11, §13.8
BA-15	RTL rules for S0, S7, S8, S9	Mirroring table and per-screen rules; RTL checklist.	§15
BA-16	Context-bar situations: Georeference, GPS follow	Add point · Delete point · Accept; Stop following.	§10
BA-17	Pairing dialogs: device name, manual code, network-blocks help	Approve shows device name; manual-code variant; hotspot hint.	§13.8

Problems A–H from the design review

	Problem	Resolution
A	Outdated Blueprint	All 17 amendments applied (Appendix A). Blueprint is superseded by this guide.
B	Overloaded top bar	Three width tiers with a fixed collapse order; readout leaves the bar on phones; Project menu absorbs Export/Import/Backup/Lock/Settings (§7).
C	Portrait space	Top 48 + dock 56 → 536 dp of map on a 640 dp window (was 472 with a context bar). Context bar only while a tool or selection is active (§14).
D	Hidden core tools	Five-slot dock: Select · Pan · Draw · Layers · More. Marker, Label, Symbols, Image, Scenes, Maps in More; Pan also by two-finger gesture (§7).
E	Context-bar conflicts	Drawing: Finish · Cancel · Undo point + one Style control on phones. Delete is always pinned directly before the final slot (§10).
F	Contrast	Recomputed; red and blue only as fills with white text; separate lighter text tints on navy and darker tints on light (§3).
G	Missing layouts	Georeference, Export wizard, GIF controls, Locate states, coordinate switcher, remote toggle + QR, location toggle all specified (§9–§13).
H	Overcrowded new screens	Georeference: side-by-side on tablet, tabs on phone. Export: four steps in one dialog; fullscreen sheet on phones (§13).

Source note. The brief says the specification defines "12 screens" but lists 11 lines. The specification defines S0–S11 = 12 screens or flows (S0 Lock/Login … S11 Pairing); this guide covers all twelve. The brief's grouping of "Presenter / Audience / Phone Remote" as Editor modes is mapped to the specification's S4, S5 and S6.

1 Overall design philosophy

MapX should feel like a broadcast control console: dense, exact and calm. The map is the canvas; everything else is a precisely placed instrument around it.

Principles

- Map first. Chrome is fixed-size sibling regions; the map receives what is left and is never covered except by the portrait sheet, map controls and transient handles (Laws L1–L7).
- Console, not consumer. Square 2–4 dp radius tiles, 2–4 dp gaps, one interaction blue, red only for danger, errors and LIVE. No gradients, no cyan/teal.
- Honest states. Every list and panel has loading, empty, error and disabled states. Imports never block with a full-screen spinner; lossy exports end with a report.
- Tablet landscape is the workstation. Phone layouts are designed for the phone, not scaled-down tablets; the phone as remote is presentation-only.
- Two themes, one identity. Dark navy and cool light-grey surfaces share the same blue, geometry and layout. The basemap style is independent of the UI theme.
- Direction-agnostic. All geometry uses start/end. Arabic RTL is a first-class mirror, tested at every stress size.
- Motion confirms, never decorates. Fades and slides of 100–240 ms confirm state changes and honour reduced motion.

What changes from Blueprint v1

Area	v1	v1.1 (this guide)
Phone dock	Select, Draw, Symbols, Layers, More	Select, Pan, Draw, Layers, More
Phone portrait bars	Top 56 · dock 64 · context 48	Top 48 · dock 56 · context 48 (only when active)
Rail item (phone)	44 dp	48 dp
Top bar	Ten controls in one row	Three tiers, explicit collapse order, Project menu
Remote	Undo, Redo, Tool	Presentation commands only
Screens	11 views	+ Lock, Settings, Accounts, Georeference, Export/Import, Pairing
Themes	Dark only	Dark and light, user-switchable
Motion	Unspecified	Off · Minimal · Smooth

Map-share budget (map area ÷ window height). Tablet landscape ≈ 82% (60% with a panel open) · Tablet portrait ≈ 83% · Phone landscape ≈ 75% · Phone portrait ≈ 83% idle, ≈ 76% while a tool is active, never below 35% of the window with a half sheet.

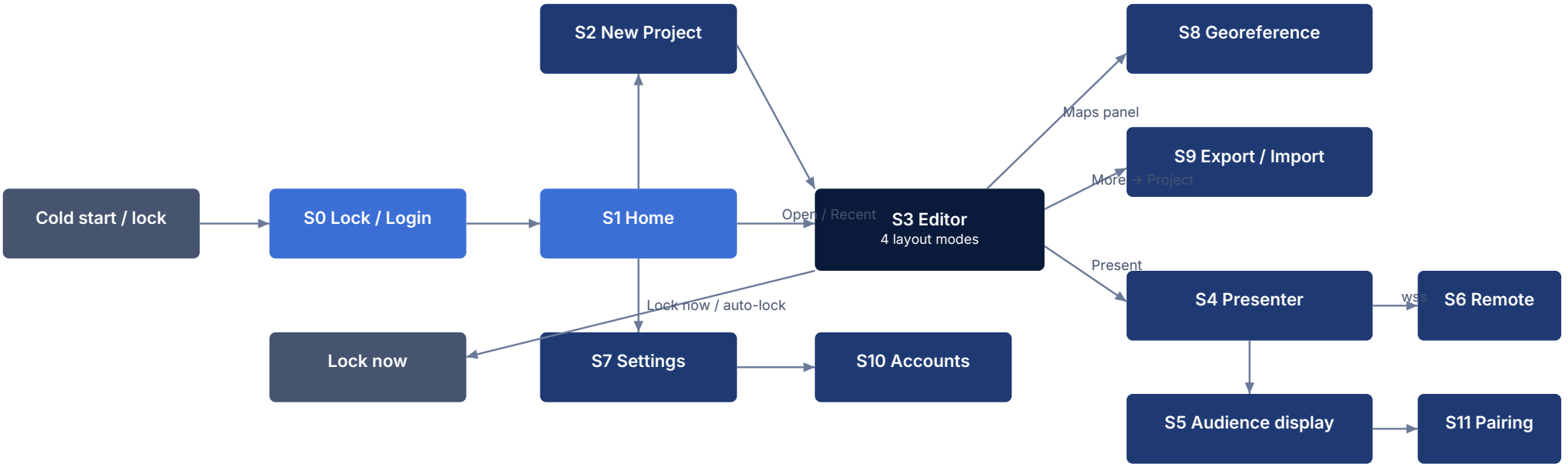
2 Overall screen architecture

Twelve screens and flows share one layout engine: one root column, a classifier that reads the window, and region siblings that never overlap.

2.1 Screen inventory

ID	Screen	Layout	Regions present	Spec / BA
S0	Lock / Login (first-run create)	Centred card (tablet) · full width (phone)	None: no rail, dock, panels, context bar; excluded from screenshots	FR-SEC-01/02 · BA-02
S1	Home	Action column + card grid (tablet) · list (phone)	Header only; no rail, panels or context bar	FR-PRJ-01 · BA-11
S2	New Project (2 steps)	Dialog (360 dp / width − 32)	ModalLayer	FR-PRJ-02
S3	Editor (four modes)	Top bar · rail or dock · map · panel/sheet · context bar	All, per mode	BA-01,05–08,16
S4	Presenter Mode	Map + graphics zones Z1–Z5 + transient bar	No chrome	FR-PRS · BA-13/14
S5	Audience screen / in-app preview	Own MapView, zones by percentage	No chrome, no controls	FR-DSP · BA-04
S6	Phone remote controller	Single scrolling column · two columns in landscape	No rail, panels or map editor	FR-REM · BA-03
S7	Settings	List + detail (tablet) · single column (phone)	Top bar only	FR-SET-02 · BA-12
S8	Georeference	Image | map | table (tablet) · tabs (phone)	Top bar, context bar	FR-GRF-06 · BA-09/16
S9	Export / Import wizards, report	One ModalLayer dialog, steps inside	ModalLayer + snackbar	FR-EXP · BA-10
S10	Accounts	Inside Settings → Security; dialogs	ModalLayer	FR-SEC-03/08 · BA-02/12
S11	Pairing flows	Dialogs (tablet) · scanner (phone)	ModalLayer	FR-REM-08 · BA-17

2.2 Navigation structure



Back: dialog/popup → panel/sheet → tool/selection → leave editor (autosave). Presenter: Back asks for confirmation. Auto-lock never fires while presenting.

Principle	Rule
Single activity	One Compose activity; screens are destinations. Dialogs and popups live in the ModalLayer; never stacked.
Back order	Dialog/popup → panel or sheet → active tool or selection → leave editor after autosave. Home → Back leaves the app.
Presenter	Back asks for confirmation; exit also by a deliberate long-press on the transient bar. Editor state restored exactly.
Lock	Every cold start, every lock and every auto-lock lands on S0; nothing from projects is visible before login.

2.3 One layout engine: classifier, region tree and the twelve laws



- L1 One root, one tree
- L2 Regions are siblings, not layers
- L3 One owner per region
- L4 Only listed overlays: modal, handles, map controls, portrait sheet, presenter bar
- L5 Panels take space (never float over the map)
- L6 One panel at a time
- L7 Sheets cover the map only
- L8 Toolbars scroll or overflow, never wrap
- L9 dp tokens from one LayoutSpec; targets ≥ 48 dp
- L10 start/end only; mirror in RTL
- L11 Classify on the window; insets once
- L12 Presenter and audience: no chrome

Classifier. Smallest window side ≥ 600 dp = TABLET else PHONE; width > height = LANDSCAPE. Re-evaluated on rotation, split-screen, resize and font scale. A tablet in split-screen below 600 dp is a PHONE layout.

3 Colour system

The settled palette is kept: navy base, interaction blue, alert red, amber marker. What changes is how it is used, because blue and red fail as text on navy.



Rule	Detail
Red is reserved	Alert red only for warnings, errors, destructive actions and the LIVE chip. Destructive buttons are red outline with text-red; only LIVE and confirm-delete use a red fill.
#5E574D is a surface	The spec tile colour is used at ~80% as a tile fill (white text 8.7:1). It is never a text colour (2.41:1 on navy).
Active = blue tile + filled icon	Active tool, active tab segment, selected row. Light theme uses the same blue, so identity does not change with the theme.

3.1 Contrast: recalculated (WCAG 2.x relative luminance)

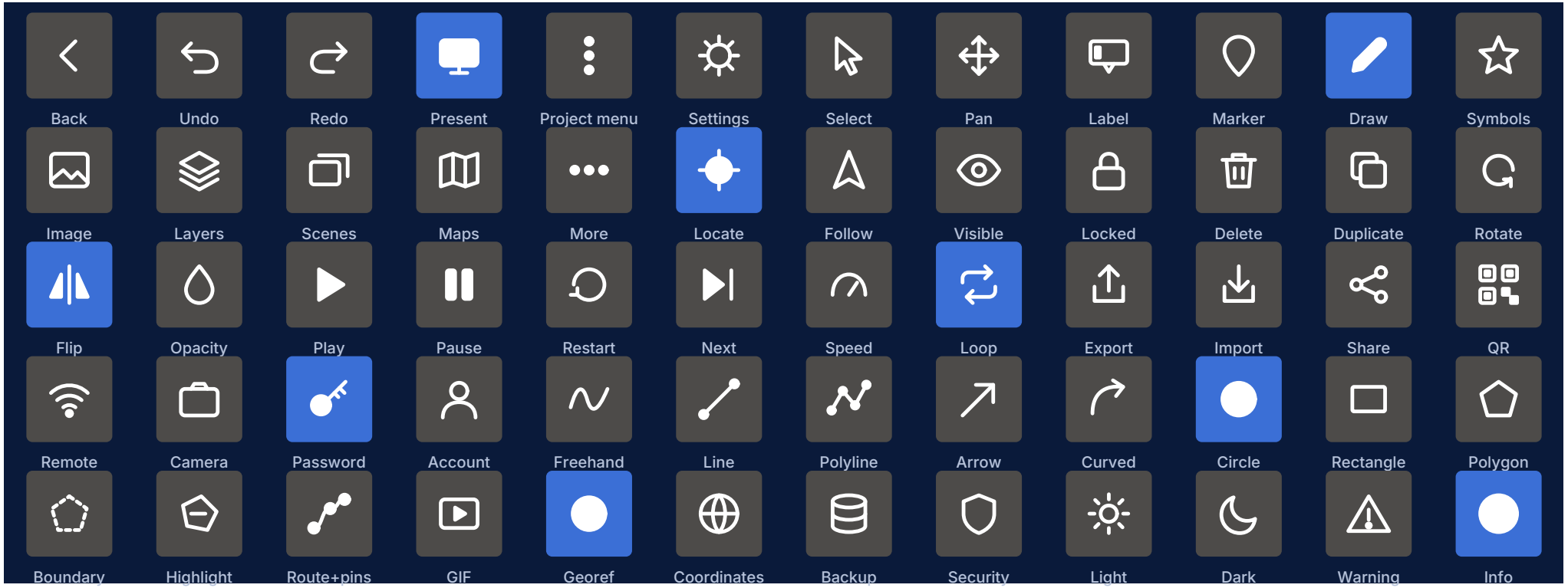
Pair (foreground on background)	Ratio	Need	Verdict
Brief: red #E83218 on navy (text)	4.01:1	4.5:1	FAIL – do not use
Spec: red #E3201B on navy (text)	3.67:1	4.5:1	FAIL – do not use
Blue #3B6FD6 on navy (text)	3.63:1	4.5:1	FAIL – do not use
Tile #5E574D (100%) on navy (text)	2.41:1	4.5:1	FAIL – do not use
White on blue #3B6FD6 (fill)	4.74:1	4.5:1	Pass
White on red #E3201B (fill)	4.68:1	4.5:1	Pass
White on blue pressed #2F5DB8	6.22:1	4.5:1	Pass
White on tile (#5E574D @ 80% on navy = #4D4B49)	8.68:1	4.5:1	Pass
Dark: text-blue #9DBBF8 on navy	8.92:1	4.5:1	Pass
Dark: text-blue #9DBBF8 on panel #0F2147	8.20:1	4.5:1	Pass
Dark: text-red #FF9A90 on navy	8.41:1	4.5:1	Pass
Dark: text-2 #B8C4DC on panel	9.01:1	4.5:1	Pass
Dark: amber #F6B800 on navy (marker, warnings)	9.64:1	4.5:1	Pass
Dark: green #3DB57A on navy	6.62:1	3.0:1	Pass
Light: navy text on bg #EEF1F6	15.18:1	4.5:1	Pass
Light: text-blue #2A55B0 on bar #E3E8F0	5.64:1	4.5:1	Pass
Light: text-red #B3170F on bg	6.09:1	4.5:1	Pass
Light: text-2 #46546F on tile #D3DAE7	5.42:1	4.5:1	Pass
Light: blue #3B6FD6 as text on bg (rejected)	4.19:1	4.5:1	FAIL – do not use
Tag navy #14264F on white (Label Tag)	14.79:1	4.5:1	Pass

Margins are thin on fills. White on #3B6FD6 is 4.74:1 and on #E3201B 4.68:1: pass for normal text but with almost no headroom. Rules: never lower the fill's lightness, never use alpha-blended blue/red as a text background, and keep button labels ≥ 12 sp SemiBold. The pressed fills darken, so contrast increases. Large-text (≥ 18.66 px bold) and non-text UI parts (outlines, icons) need 3:1 and all pass.

Semantic mapping. Connected = green dot + word; Reconnecting = amber dot + word; Disconnected = grey dot + word; Session expired = red fill chip with white text. Colour is never the only carrier: every status has a word or icon.

4 Icon system

Outline icons by default; the same glyph switches to filled when its tool or panel is active. A 24 dp glyph on a 24 × 24 grid with a 1.8 dp stroke, round caps and joins.



Rules

- Sizes: glyph 24 dp (18 dp inside chips, 20 dp in rows, 22 dp in menus); touch target always ≥ 48 dp even when the visible tile is smaller.
- States: inactive = outline in text colour; active = filled glyph on blue tile with white glyph; disabled = text-disabled; destructive = text-red outline.
- Small-size clarity: no detail finer than 1.5 dp; interior marks (pin hole, label square) are knocked out with the tile colour when filled.
- Every icon-only control has a content description and a long-press tooltip; phone context bars rely on this.
- Distinct overflow icons: vertical kebab = Project menu (top bar); horizontal dots on a labelled dock tile = More tools.

Direction-aware icons (RTL)

Flip in RTL	Do not flip
Back, Next / Previous, skip icons, Undo, Redo, progress direction, import/export arrows, convert, list chevrons	Play/Pause, clocks, compass and north arrow, the map itself, numbers, Latin brand names, Locate, Layers, Select cursor (mirrors only via text position)

Active-state recipe

Rail tile 56 × 56 (tablet) / 48 × 48 (phone landscape) / dock slot ≈ 78 × 48: radius 3 dp, fill blue, glyph filled white, label white 11 sp Medium. Inactive: fill tile, outline glyph in text colour.

5 Typography

Cairo (SIL OFL, bundled) for Arabic and Latin; works offline. Numbers always use Western digits by default and keep their reading order in RTL.

Role	sp / line	Weight	Used for
Display	24 / 32	Bold	Lock title, empty states
Title	18 / 24	SemiBold	Screen titles, Home headings
Panel title	15 / 20	SemiBold	Panel/sheet header, project name in top bar
Body	14 / 20	Medium	Rows, menus, dialog body
Field value	13.5 / 18	Medium	Properties fields
Button / chip	12.5–13 / 16	SemiBold	Buttons, context chips
Caption	11 / 14	Medium	Field labels, rail/dock labels, helper text
Micro (min)	10.5 / 13	Regular	Row sub-lines, attribution only
Coordinates	12 / 16	Medium, tabular	Readout, properties, import preview

Rules

- Minimum 10.5 sp, only for sub-lines; interactive labels ≥ 11 sp. All text scales with system font size to 200%: bars keep their height class and truncate or scroll; panel bodies wrap and scroll.
- Number styling: tabular lining figures; coordinates, zoom, scale and counts are isolated as LTR runs (Unicode LRI) so "35.1234° N 47.8821° E" never reorders in Arabic. Hemisphere letters N/E/S/W stay Latin in both languages (FR-CRD-06).
- Digits: Western by default; optional Arabic-Indic digits setting (P2) affects display only; input accepts both.
- Arabic: Cairo has ~10–15% taller ascenders; line heights above already include it. Arabic is never letter-spaced and never faux-bolded; use the real weights.
- Label Tag (map text): bold, navy #14264F on white; sizes S/M/L = 11/12/14 sp; platform text layout with Cairo, so shaping matches the UI.



Type specimen, LTR and RTL

6 Interaction and motion

Smooth by default; three user levels. Motion only confirms state changes and always yields to reduced-motion.

Animation setting (Settings → Appearance)

Level	Behaviour
Off	No transitions; state changes are instant. Live map movement (user gestures) is unaffected.
Minimal	Fades only, 100 ms. Sheets and panels appear without slide. Scene fly-to becomes a cross-fade.
Smooth (default)	Fades 120 ms; panels and sheets slide 200–240 ms (standard ease-out); selection handles scale-in 120 ms; toggles 120 ms.

Android's system "Remove animations" / reduced-motion forces Off regardless of the setting. Presenter transitions (instant, fly-to, fade, object appearance) follow the same rule (FR-PRS-07, FR-ANI-04).

Duration tokens (ms)

Token	ms	Use
motion.fast	100	Press, toggle, chip state
motion.base	120–160	Fades, handles, popups
motion.panel	200–240	Panel/sheet slide; dialog scale 0.96→1
motion.scene	≤ 800	Fly-to (presenter only)

Gestures

Gesture	Result
One-finger drag (Pan tool)	Moves the map. In Select, a drag on empty map also pans; a drag on an object moves it.
Two-finger drag (any tool)	Pans the map without leaving the active tool (so Pan is optional but always a tool).
Pinch / two-finger rotate / two-finger tilt	Zoom, bearing, pitch. Double-tap zooms in; two-finger tap zooms out.
Tap	Select (Select tool), place (Label, Marker, Symbol), add vertex (drawing).
Long-press on empty map (Select)	Creates a Label at that point.
Drag a handle	48 dp hit area around 12 dp visible handles; one gesture = one undo step.
Stylus	Draws and drags handles; palm rejection by platform stylus events.
Long-press an icon-only control	Tooltip with its name.

Feedback

- Press: tile darkens to blue-pressed / tile-pressed in 100 ms; Android ripple is disabled on square tiles in favour of a flat state change.
- Selection: 2 dp blue outline + handles; a changed object flashes once (Smooth only).
- Save status: dot colour change with the word in the top bar; Error persists until resolved.
- Snackbars 4 s, one at a time, above the context bar.

7 Main application navigation

Top bar, tool rail / dock and the overflow menus. One owner per region; overflow is explicit and ordered, never improvised.

7.1 Top bar: width tiers and collapse order (problem B)



Tablet 1280 dp: all controls visible; coordinate readout centred and tappable



Phone 390 dp: Back · name + save dot · Undo · Redo · Present (icon) · Project menu



Phone 360 dp: same items; name area 112 dp (truncates with ellipsis)

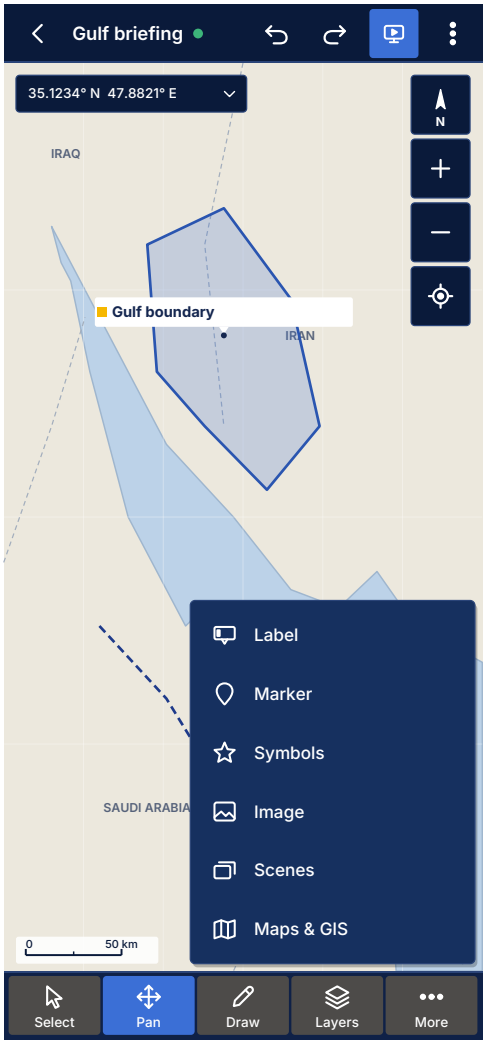
Item	Tablet (≥ 600 dp)	Phone	Narrow (< 340 dp)
[redacted]	[redacted]	[redacted]	con 48
Project name	Visible, tap to rename (max 170 dp)	Ellipsis; tap to rename	Ellipsis
[redacted]	[redacted] Unsaved / Error)	[redacted] word in a tooltip	Dot
Coordinates · zoom · scale	Centre chip, tappable → format popup	Map-corner chip (start-top), tappable	Same
Undo · Redo	Icons	Icons	Project menu
Present	Blue button with word	Blue icon tile 40 dp (hit 48)	Same
Settings	Gear icon	Project menu → App group	Same
Project menu ( )	Icon: Project group	Icon: Project + View + App groups	Same

very small windows: Undo/Redo move into the Project menu (Redo first)

- Collapse order (when width runs short): readout leaves the bar → Present label becomes icon → Undo/Redo go to the Project menu → name keeps ≥ 64 dp. The name truncates last.
- Every item keeps a 48 dp target; the bar height never changes with font scale; nothing wraps.
- Phone widths: 48 Back + 4 × 48 = 240 dp of controls leave 150 dp (390) or 120 dp (360) for the name and dot.
- LIVE: while presenting externally a red LIVE chip (80 × 32) replaces the readout on tablets; on phones it replaces the name area.

> Project menu content (BA-05). Project group: Export..., Import..., Backup..., Project properties, Lock now window height and scrolls.

7.2 Tool rail, five-slot dock and the More menu (problem D)



Phone portrait: dock = Select · Pan · Draw · Layers · More

Slot	Opens	Notes
Select	Tool	Default tool. Tap selects, drag moves, long-press on empty map creates a label.
Pan	Tool	Map gestures only; separate from Draw. Two-finger drag pans in every tool.
Draw	Popup (ModalLayer)	11 tools. Chosen tool shows in the context bar; the slot stays blue while a drawing tool is active.
Layers	Layers sheet	Tap again to close. Sheet peek 72 · half 50% · full.
More	Popup	Label, Marker, Symbols, Image, Scenes, Maps & GIS. Lists only items not visible in the current bar.

Mode	Bar	Items
Tablet L/P	Rail 72 wide, tiles 56 × 56, gap 4	Select, Pan, Label, Marker, Draw, Symbols, Image, Layers, Scenes, Maps (scrolls vertically)
Phone landscape	Rail 56 wide, items 48 × 48, gap 4	Select, Pan, Draw, Layers, Scenes, More (Label, Marker, Symbols, Image, Maps & GIS)
Phone portrait	Dock 56 high, 5 slots	Select, Pan, Draw, Layers, More (Label, Marker, Symbols, Image, Scenes, Maps & GIS)

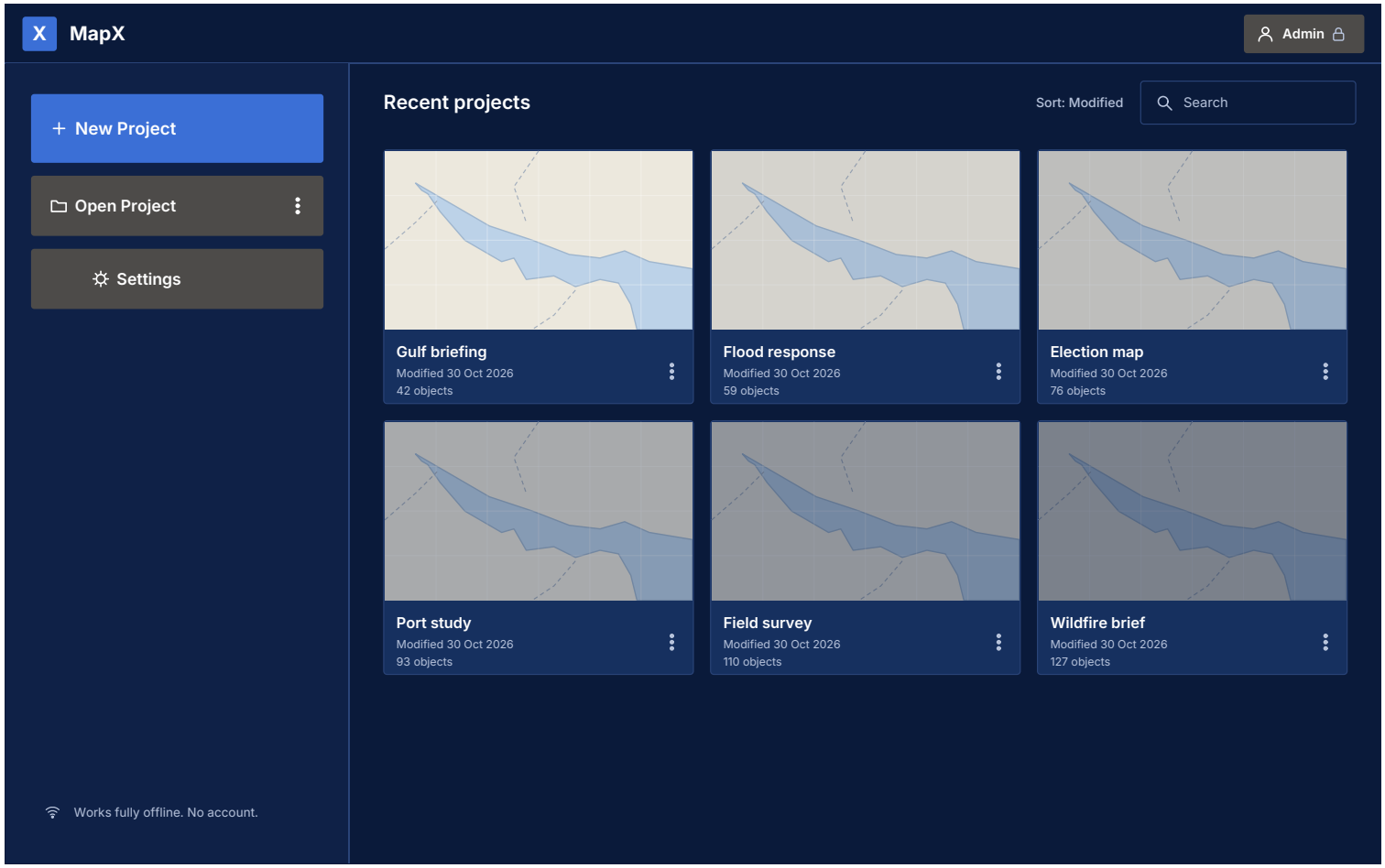
Tablet shows all ten; Marker, Label and Pan are never hidden on tablets. On phones Label and Marker sit one tap away in More and remain one-tap place to
Cancel).

7.3 Panel and sheet rules (PanelManager)

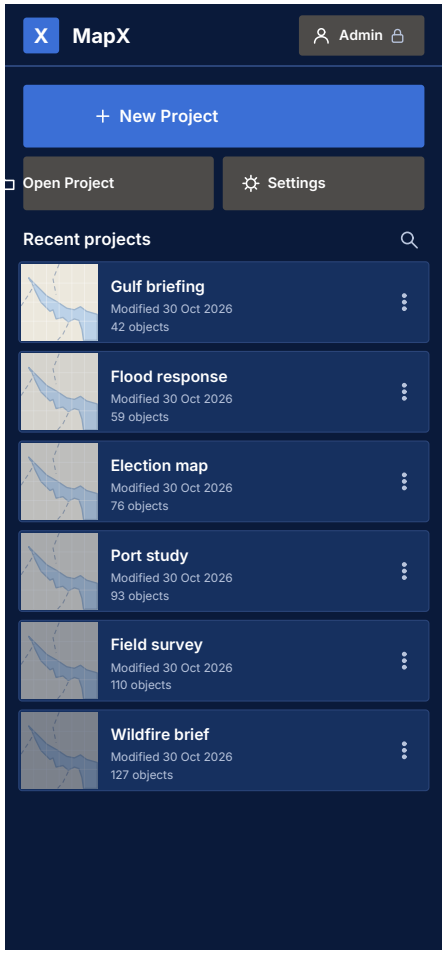
Rule	Behaviour
One at a time	ActivePanel {None, Properties, Layers, Symbols, Scenes, MapSources, GisImport, SceneGraphics}. Opening another replaces it in the same slot.
Open / close	Tap the rail/dock item to open; tap again or the to close. Android Back closes it before cancelling the tool.
Selection	Selecting opens Properties only if the active panel is None or Properties; otherwise the panel stays and the context bar offers [Properties].
Shape	Tablet / phone landscape: side panel, fixed 320 dp (280 on phone landscape) at the end edge, with a close button; the map narrows. Portrait: bottom sheet over the map only.
Structure	Header 44 dp (title, optional overflow, close) · scrolling body · fixed footer 48 dp. Sheet adds a 16 dp drag handle; peek (72 phone / 80 tablet) shows handle + header only.
Rotation / resize	Same panel stays open and re-composes between panel and sheet without losing scroll position or field input.
Presenter	All panels close and are disabled.

8 Home screen

Adaptive: an action column and a card grid on tablets, a single list on phones. Home has no rail, no panels and no context bar (FR-PRJ-01).



Tablet: 320 dp action column · 3-column card grid

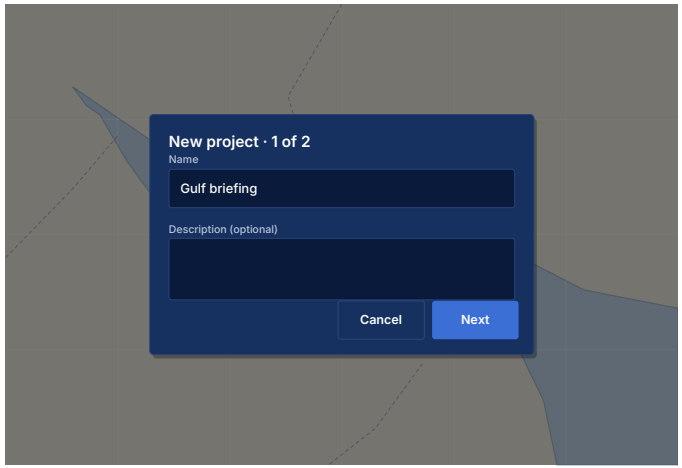


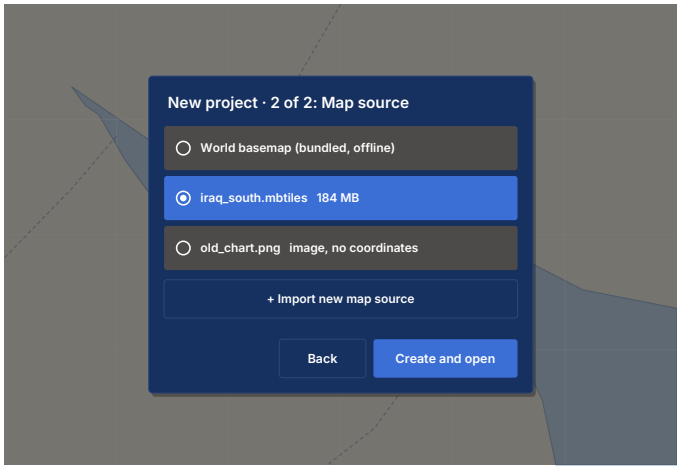
Phone: actions first, recents list

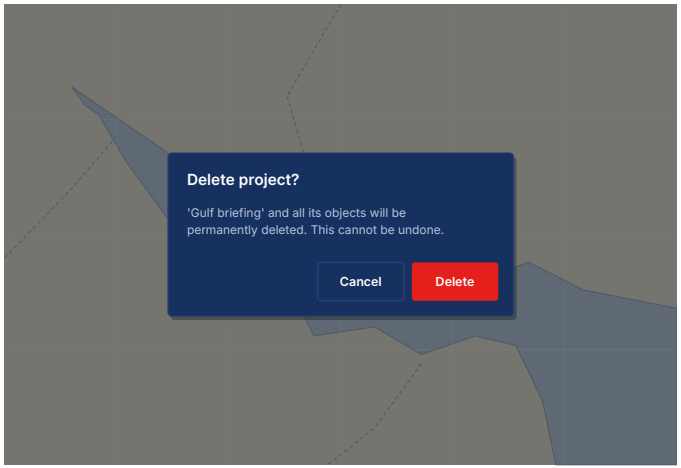
Element	Spec
Header	56 dp. Logo tile 32 dp · account chip (name + lock, 112 × 36, hit 48) → menu: Lock, Switch account, Settings (BA-11).
Actions	New Project (primary blue, 64 dp) · Open Project (56 dp; overflow adds Import project package…) · Settings. Footer note: “Works fully offline. No account.”
Cards / list rows	Thumbnail, name, modified date, object count, overflow (Rename, Duplicate, Delete). Duplicate = deep copy with new ids. Delete is always confirmed.

Element	Spec
Sort / search	Search 200 dp. Sort: Modified (default), Name, Created.
States	Empty: icon + “No projects yet — New Project”. Loading: card skeletons. Error opening a project: dialog with Restore last-known-good and Back to Home. Missing map source: project opens with a warning bar and Relink.

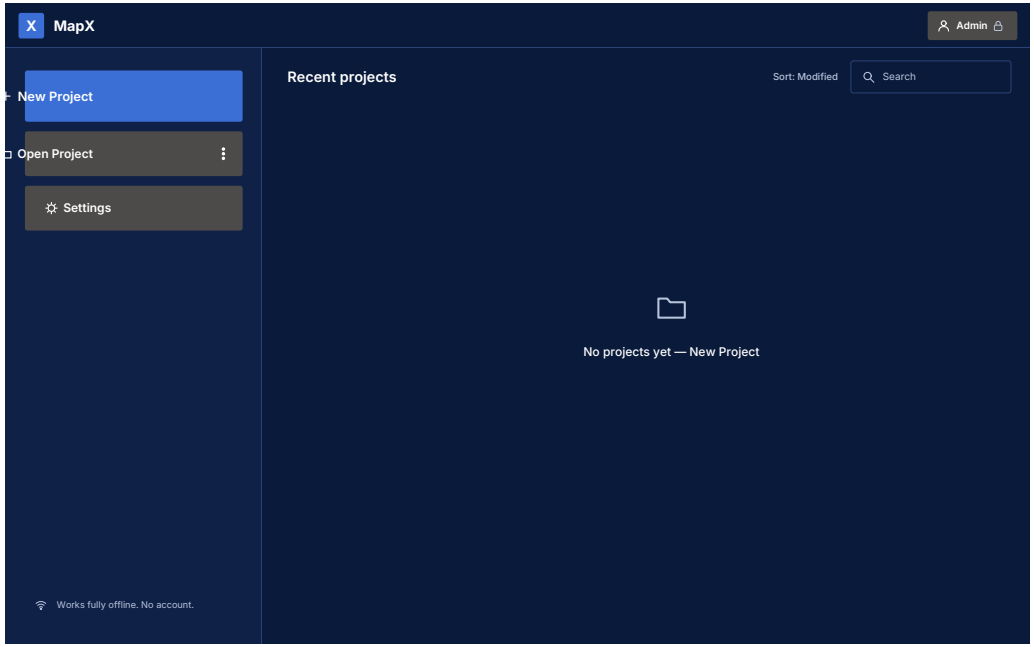
8.1 New Project, Delete, Open-menu and Home states



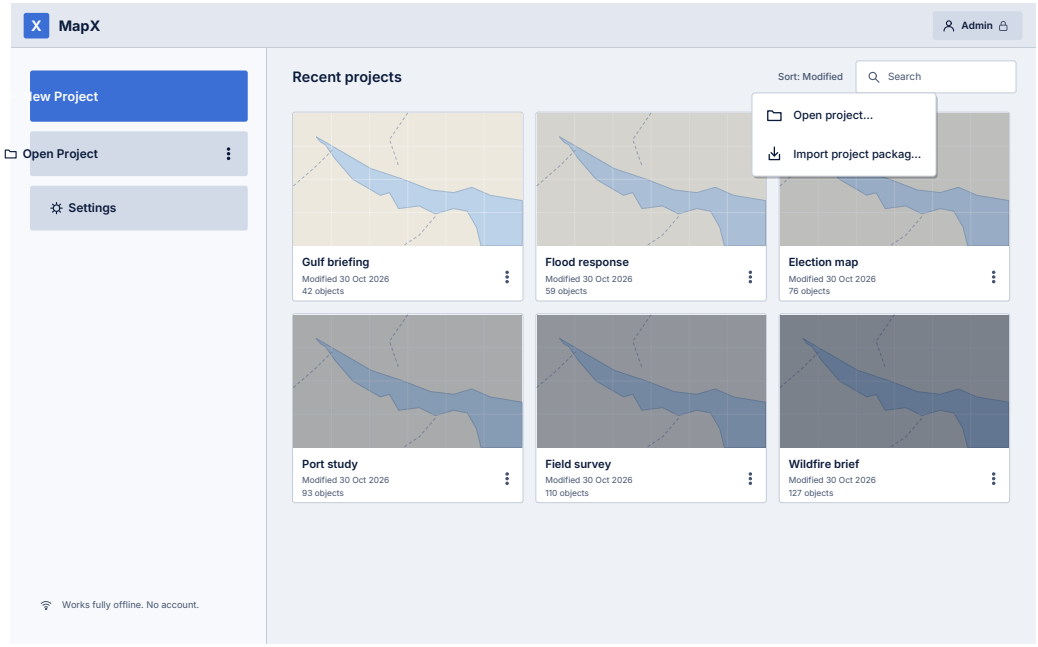




Dialogs (ModalLayer): 360 dp wide on tablet, width – 32 on phones; primary action blue at the end edge, destructive red, Cancel first; one at a time



Empty state (tablet)



Light theme + Open Project overflow (Import project package...)

9 Map interface

The map is the dominant element. Everything around it is a fixed-size sibling; the only things on top of it are its own controls, handles and, in portrait, the sheet.

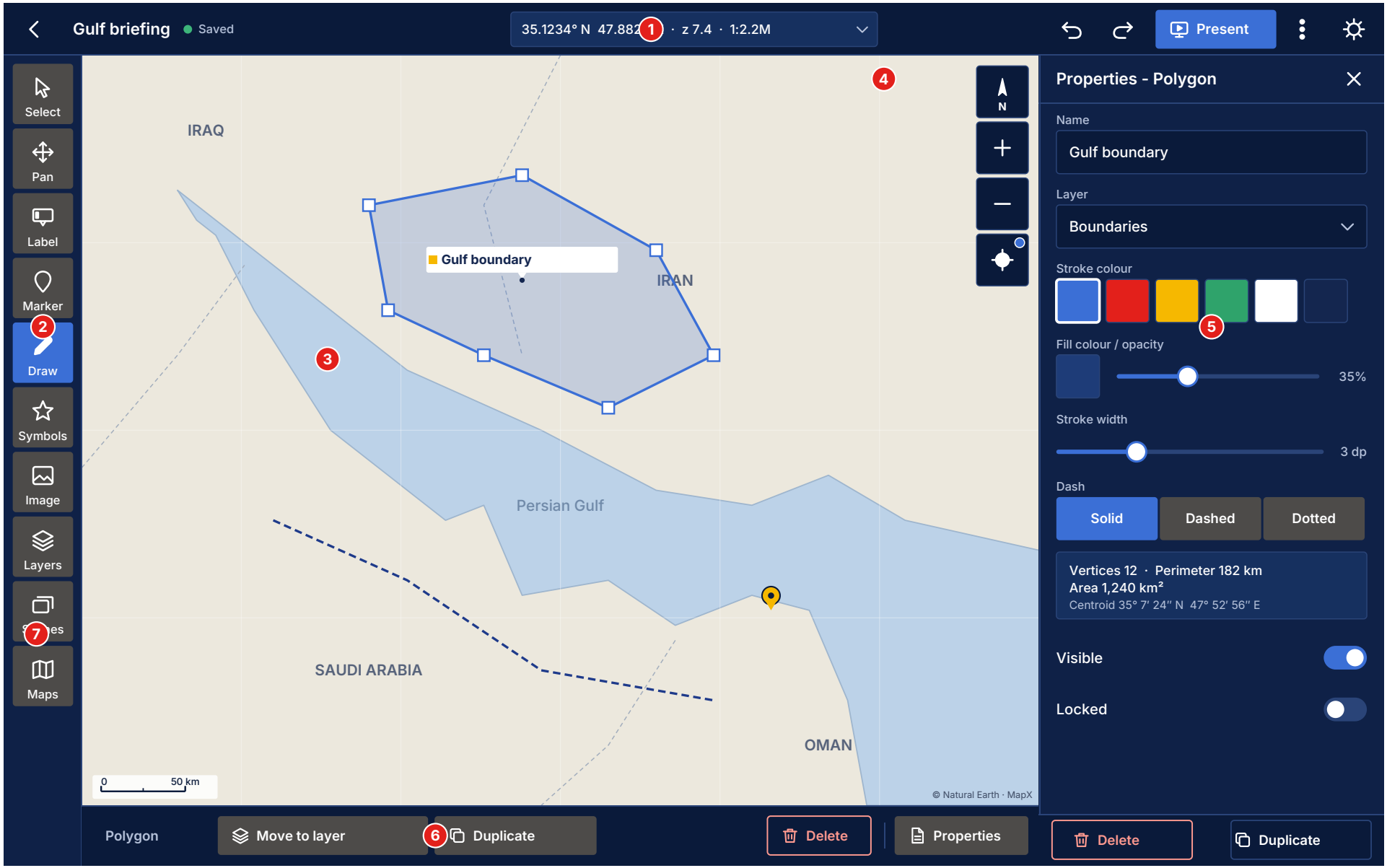


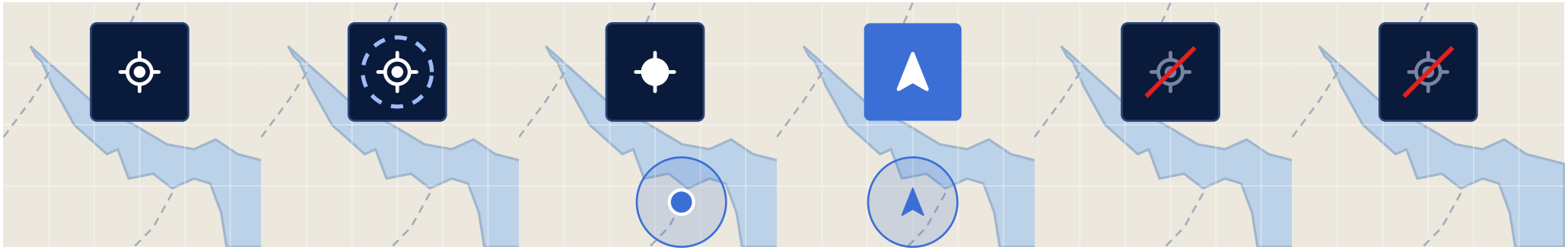
Figure 9.1 Tablet landscape (1280 × 800 dp), panel open, Draw active, polygon selected.

9.1 Regions

#	Region	Contents and v1.1 rules
1	TopStatusBar 48	Back · name + save status · readout chip (tappable) · Undo · Redo · Present · Project menu · Settings. Spans full width.
2	ToolRail 72	10 tiles 56 × 56, gap 4, 8 dp side padding; scrolls with a fade at the cut edge. Active = blue tile + filled glyph.
3	MapWorkspace	MapLibre map, objects, selection outline and handles, in-progress drawing. ~82% of the window (60% with a panel).
4	Map controls	Vertical stack at the end edge, 10 dp inset: compass/north → zoom in → zoom out → Locate. Tiles 48 × 48, gap 4. Scale bar at the start-bottom, attribution at the end-bottom.
5	SidePanel 320	Header 44 + close · scrolling body · fixed footer 48. Min 280, max 360, fixed 320 by default; does not grow with font scale.
6	BottomContextBar 56	Actions of the active tool/selection; scrolls horizontally, never wraps; pinned end slots (Delete, Properties).
7	ModalLayer	Dialogs, menus, popups, snackbar; one at a time; above everything.

Selection and handles. Selected objects get a 2 dp blue outline and handles: 12 dp visible, 48 dp hit area, transient and never stored. Multi-select shows one outline per object. Imported (read-only) objects show a lock badge and no handles.

9.2 Map controls, Locate states and follow mode (BA-06)



Idle Tap: asks once, one-line rationale first

Searching Dashed ring; tap cancels

Located Dot + accuracy circle; tap centres/fits

Follow Blue tile; camera follows; shows Stop fo

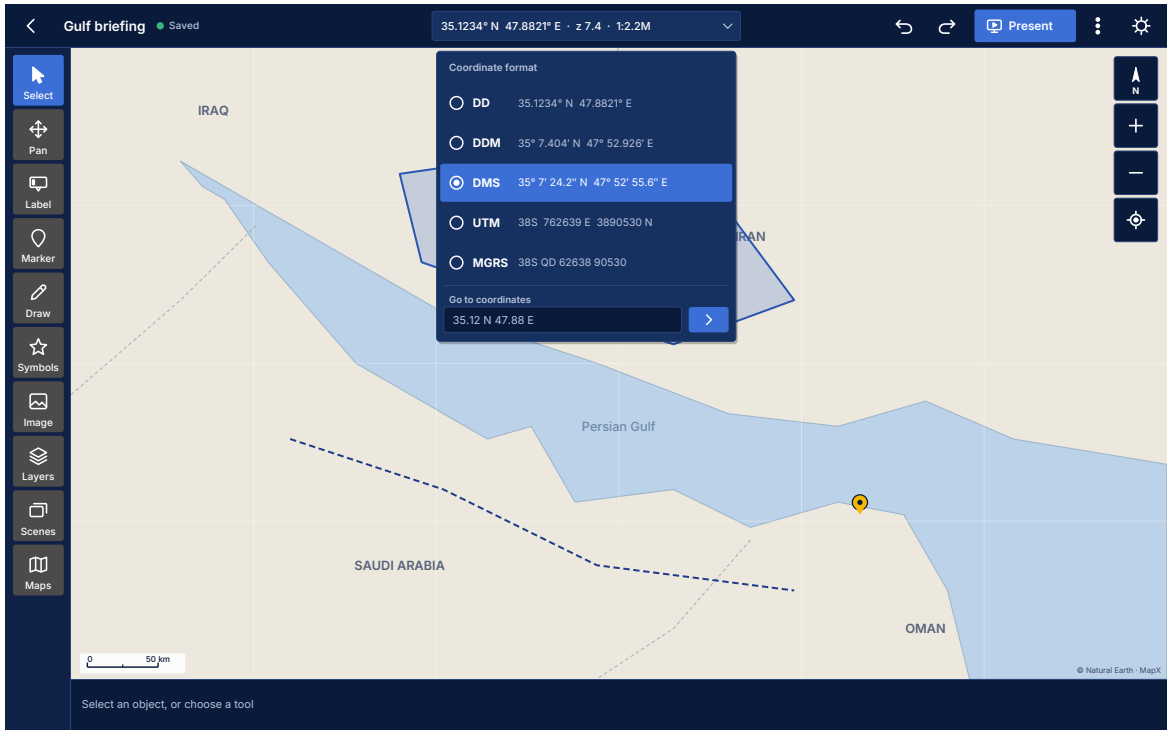
Denied / off Struck glyph; message + Open settings

No GNSS Disabled with message; app unaffected

Locate state	Behaviour and message
Idle → request	First tap shows a one-line rationale (“MapX uses your location only while this screen is open; it is never saved or sent.”) then the system permission. Foreground only.
Searching	Dashed ring on the tile; if no fix after 20 s the snackbar says “Still searching — GNSS works best outdoors.”
Located	Dot with accuracy circle; Android may grant approximate only: the accuracy circle shows it. Second tap centres/fits; long-press on the tile offers Drop marker at my location (undoable command).
Follow	Third tap (or long-press → Follow). Tile turns blue; camera follows; heading arrow when sensors allow. Context bar shows Stop following (BA-16). Any manual pan stops following.
Denied / off / no GNSS	Struck glyph. Message with a shortcut to system settings; “This device has no GNSS receiver” disables the tile. Nothing else changes.

- Stack position: compass, zoom in, zoom out, Locate, top-down from the end-edge top corner. It rides above sheet edges: if the visible map is shorter than 232 dp the stack compacts to compass + Locate (pinch still zooms).
- Compass shows bearing; tap resets north. North glyph does not flip in RTL.
- Show my location (presentation): per-presentation toggle in Scenes → Presentation options, default off; the dot never appears on the audience screen unless on (BA-13).
- Scale bar: metric only, live from latitude, 116 × 22 dp, white 88% so it reads on any map.

9.3 Coordinate readout and format switcher (BA-05, FR-MAP-04/05)



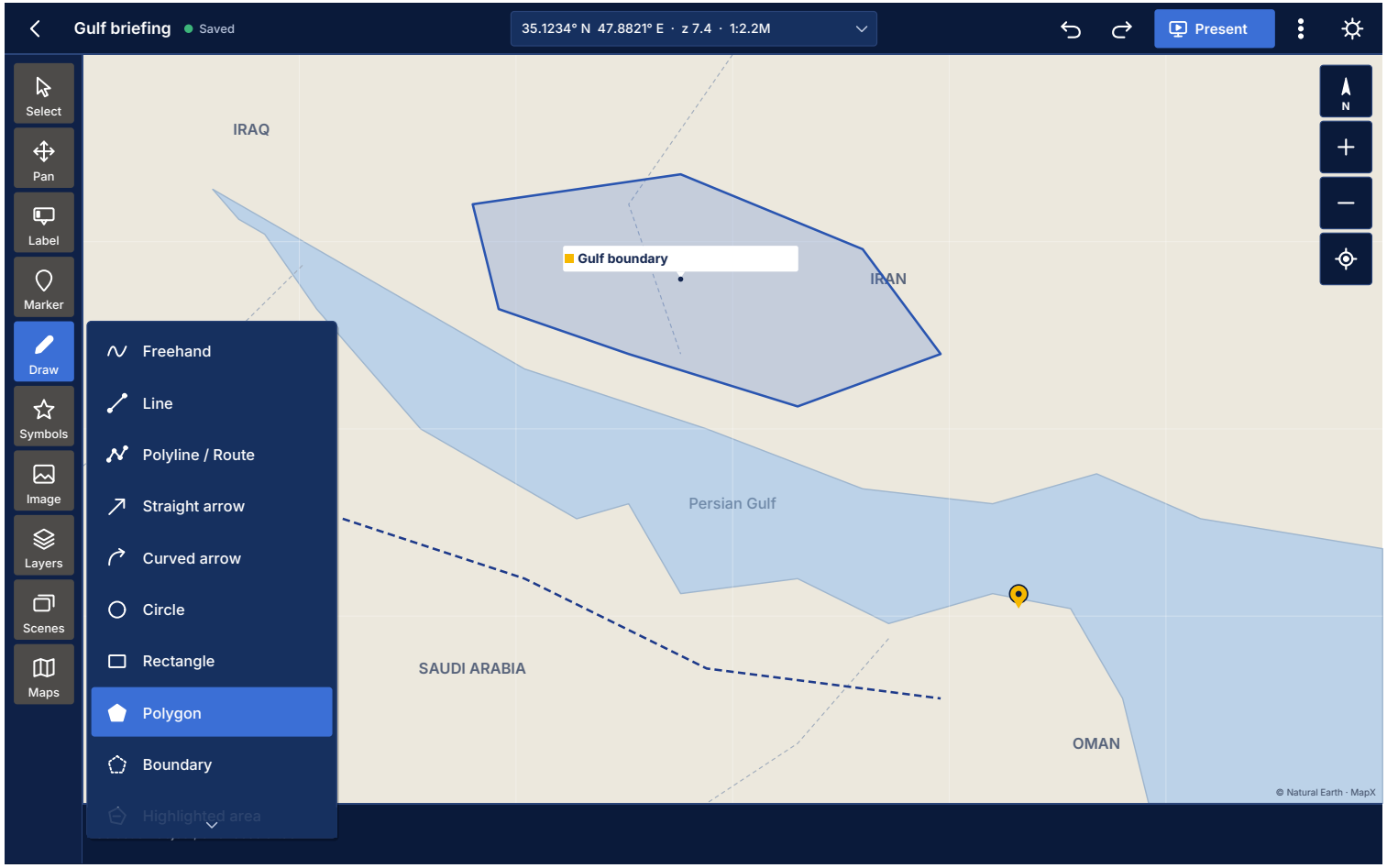
Tablet: readout chip → popup under it

- The readout (tablet: centre of the top bar; phone: 188 × 30 dp chip at the map's start-top corner) is tappable and opens the coordinate-format popup.
- Popup (330 dp, ModalLayer): radio list DD, DDM, DMS, UTM, MGRS with a live example of the current map-centre in each; the choice is saved per account and applies everywhere (status, properties, GPS, search, import preview).
- Go to coordinates (DEV-8): one field at the bottom of the popup accepts any format (auto-detected; Arabic-Indic digits accepted); invalid input shows an inline explanation. This keeps FR-MAP-05 off the top bar.
- UTM/MGRS cover 80°S–84°N; outside that band the readout shows DD and a short note.
- MGRS precision (10 km … 1 m) is chosen in Settings → Coordinate format.

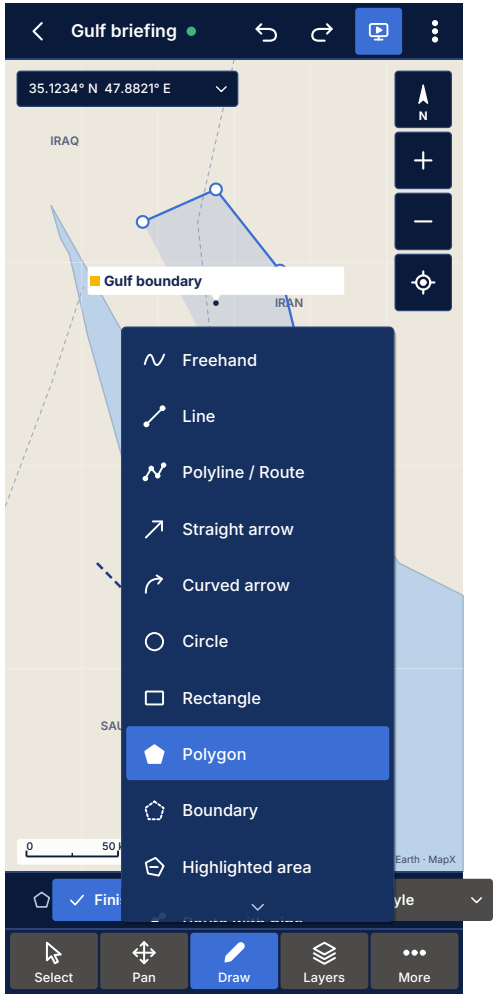
> Example values in the mock-ups are computed for 35.1234° N, 47.8821° E: UTM 38S 762639 E (AC-CRD-1).

10 Drawing interface

Eleven tools in one popup; each shows its actions in the context bar. The context bar never wraps: it scrolls horizontally and pins the last two slots.



Draw popup (tablet): 232 dp wide, 48 dp rows, max 60% height, scrolls



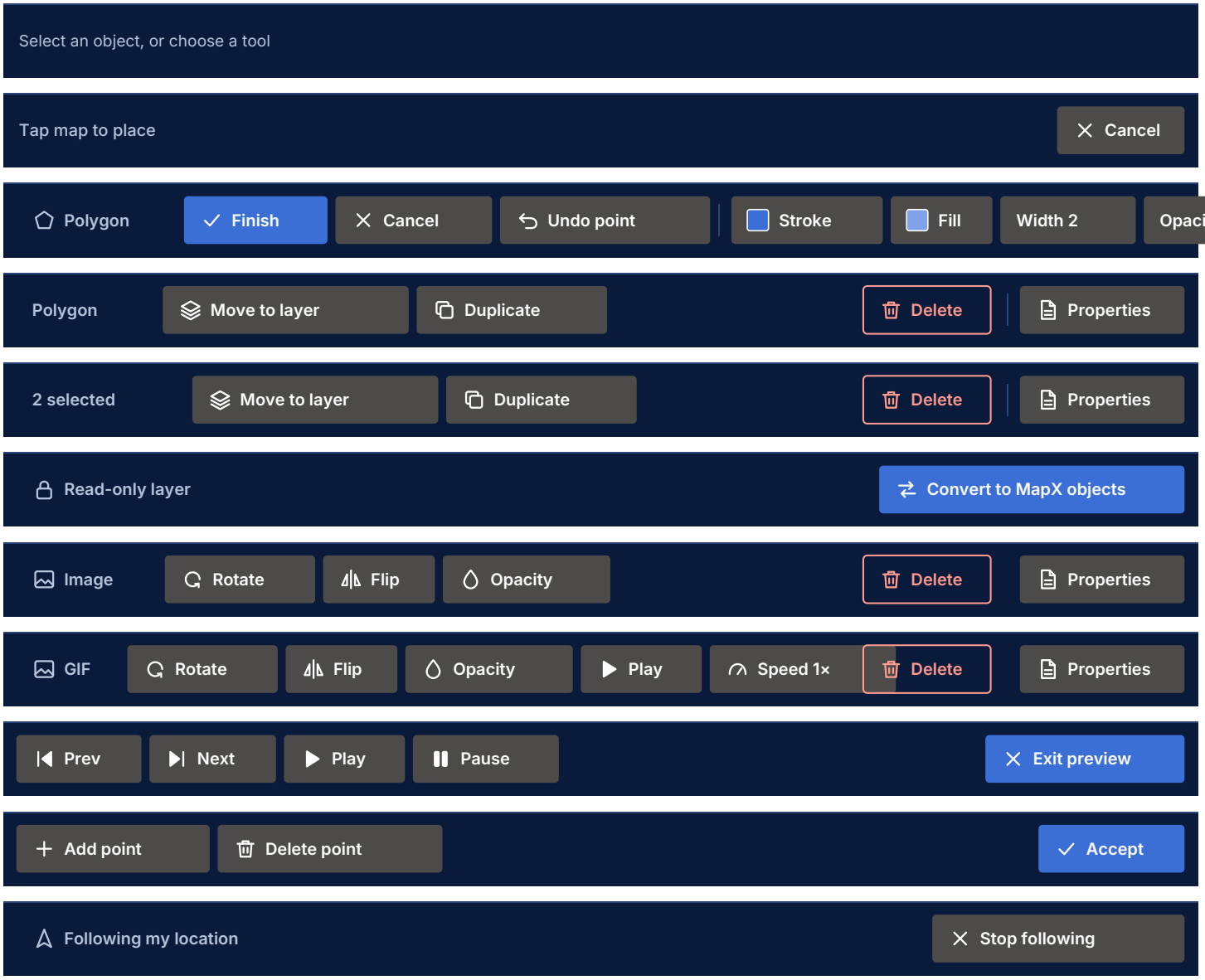
Phone: popup above the dock, drawing in progress

10.1 The eleven tools (FR-DRW-01)

Tool	Gesture	Finish	Stored geometry (all geographic)
Freehand	Press-drag-lift	On lift (one command)	Points simplified (Douglas–Peucker, ~2 dp) with a hard vertex cap
Line	Tap start, tap end	Second tap	Two points
Polyline / Route	Tap each vertex	Finish or double-tap	Ordered point list
Straight arrow	Tap start, tap end	Second tap	Two points; arrowhead sized in dp at render time
Curved arrow	Start, end, drag the control point	Release	Start, end, one geographic control point
Circle	Tap centre, drag radius	Release	Centre + radius in metres (geodesic)
Rectangle	Drag diagonal	Release	Centre + width/height in metres + bearing
Polygon	Tap vertices	Finish (≥ 3 points) or tap first vertex	Ordered ring
Boundary	Tap vertices (dashed outline)	Finish	Ordered point list
Highlighted area	Tap vertices (translucent fill)	Finish	Ordered ring
Route with pins	Tap vertices; long-press adds a pin	Finish	Line + pin points

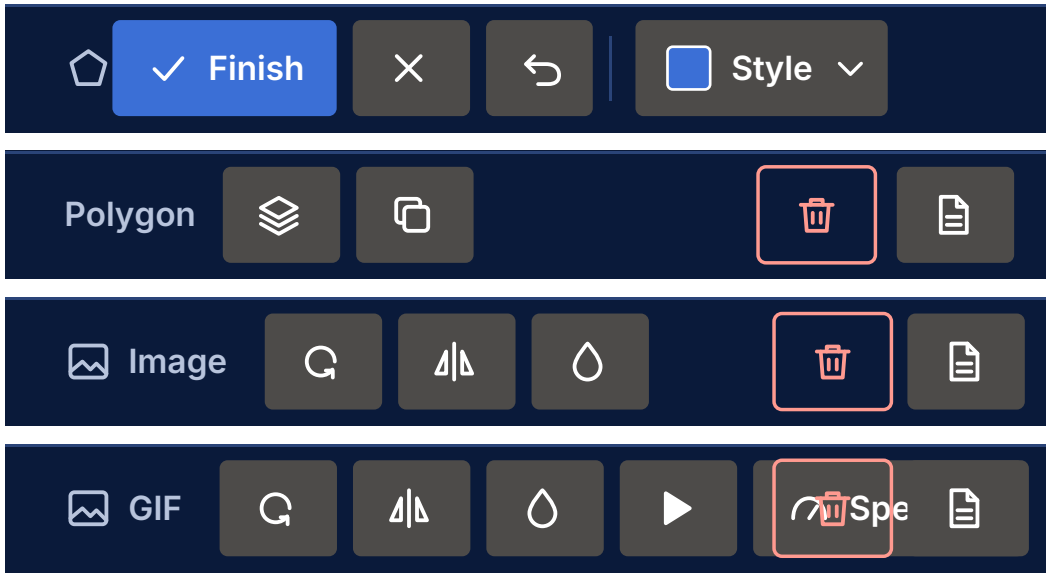
- Active drawing tool is shown in interaction blue in the rail/dock and its name appears first in the context bar.
- While drawing: first vertex gets a 12 dp target to close; the rubber band is a dashed blue line; Undo point removes the last vertex only; Cancel discards the in-progress shape; Finish commits one command and selects the result.
- Style defaults are the last-used style (stroke, fill, width, opacity, dash, arrowheads); changing style mid-draw updates the preview.
- A circle around a city stays around that city when the map pans, zooms or rotates.
- Stylus: precise vertices, handle drags and freehand; touch and stylus share the same 48 dp handle hit areas.
- After Finish the tool stays active for the next shape; tapping Select ends it.

10.2 Context bar situations (problem E, BA-16)



Conflict resolved. The earlier bar showed seven drawing actions and promised “Delete second-to-last” while Properties was last, which no situation actually satisfies (read-only objects have no Delete; images had none after it). Rule now: every bar is [info] [primary actions] [style] then two pinned end slots; Delete (red outline) always sits immediately before the final slot with a divider, and the final slot is Properties (or the primary action). Style chips (Stroke, Fill, Width, Opacity) show inline on tablets and collapse into one Style control on phones.

10.3 Phone context bars (48 dp, icon-first, scroll horizontally)



- Phone items are 44 × 36 dp visible with a 48 dp hit area; icon-only except Finish, Style and tool name; every icon has a content description and long-press tooltip.
- The pinned pair (Delete, Properties) never scrolls away; the scrolling part fades at its cut edge to hint overflow.
- The Style control opens a small popover (ModalLayer) with Stroke, Fill, Width, Opacity, Dash: 4 swatch rows and two sliders, 48 dp rows.
- Properties opens the same composable as a bottom sheet (peek 72 · half 50% · full); on tablets it is the 320 dp side panel.
- In Presenter Mode there is no context bar. In Georeference the bar shows Add point · Delete point · Accept.

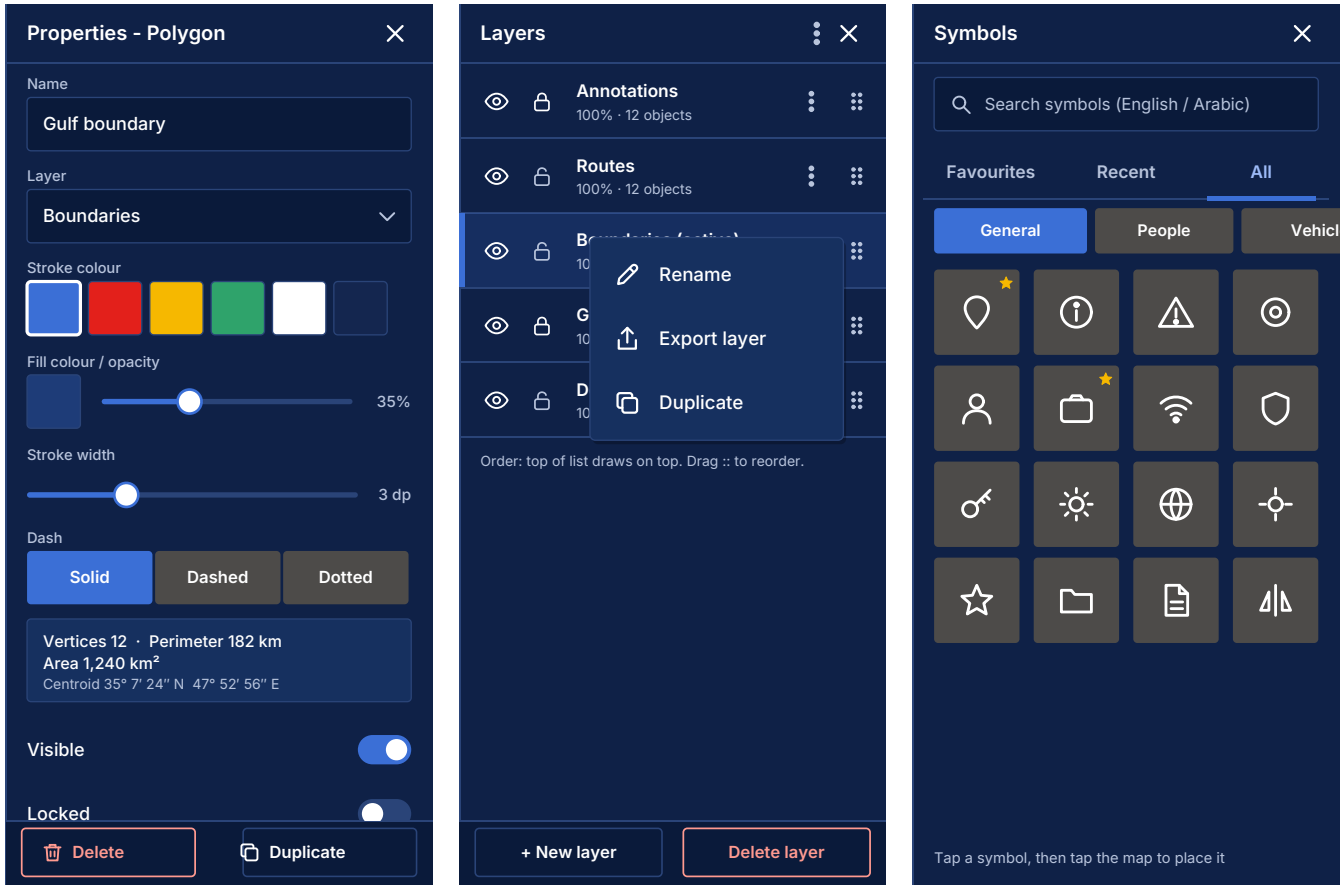
11 Tool and panel specifications

Ten tools, seven panels. All panels share one frame: header 44 dp, scrolling body, fixed 48 dp footer, 320 dp wide (280 on phone landscape; bottom sheet in portrait).

11.1 Tools

Tool	Type	Behaviour	Context bar
Select	Tool	Tap selects (blue outline), tap empty deselects, multi-select, drag moves without panning, long-press empty creates a Label.	Selection actions
Pan	Tool	Map gestures only (also available by two-finger drag in any tool).	Hidden / hint
Label	Place tool	Tap to place a Label Tag: white card, bold navy #14264F text, amber 8 × 8 dp anchor square on the inline-start edge, pointer to the coordinate. Sizes S/M/L; bidi-safe.	[Tap map to place] [Cancel]
Marker	Place tool	Tap to place a marker (pin, amber default). Properties: name, icon, colour, rotation, show-name.	[Tap map to place] [Cancel]
Draw	Popup	Eleven tools (§10).	Tool actions
Symbols	Panel	Search (English/Arabic), Favourites · Recent · All, 15 categories, grid; tap then tap map.	Place hint
Image	System picker	PNG/JPG/WebP/GIF/animated WebP → image object (§12).	Image actions
Layers	Panel	One active layer; visibility, lock, opacity, order.	—
Scenes	Panel	Scene list, add from current view, graphics, present from here, presentation options.	—
Maps	Panel (2 tabs)	Map Sources | GIS Import; import, inspect, select, delete, Georeference.	—

11.2a Properties, Layers (with row menu) and Symbols: visuals

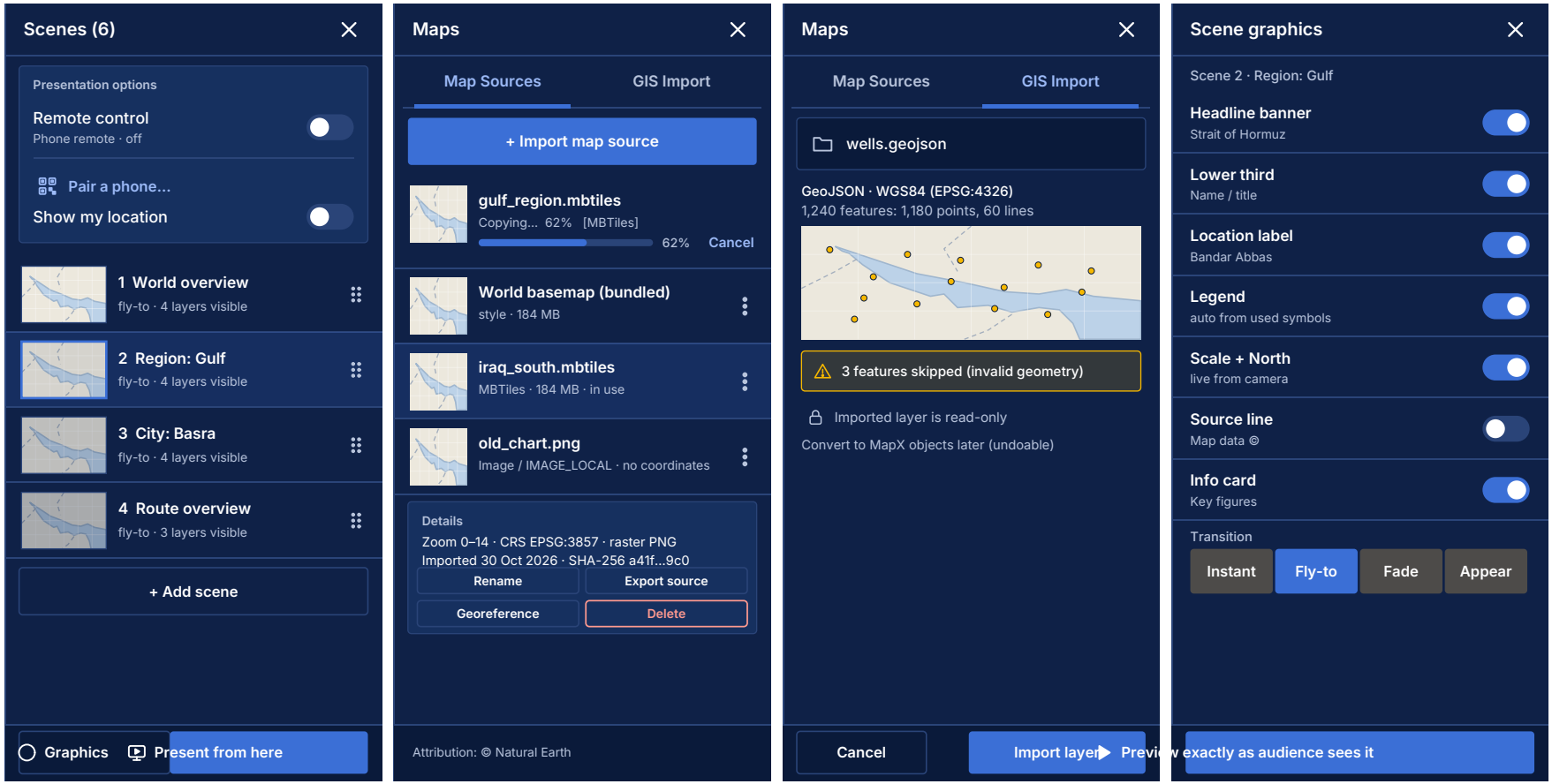


11.2 Properties, Layers and Symbols: specification

Panel	Specification
Properties (FR-OBJ-09)	One composable reused as side panel (tablet) and sheet (phone). Fields: Name, Layer (dropdown), Stroke colour (6 swatches 40 dp, 8 gap, custom picker), Fill colour/opacity, Stroke width (slider + value), Dash (Solid / Dashed / Dotted segmented 40 dp), measurements card (vertices, perimeter, area in km²/ha, centroid in the chosen coordinate format), Visible and Locked toggles. Footer: Delete (red outline) · Duplicate. Per-type editors come from MapObjectDefinition; Label adds text, size S/M/L, tag and anchor colour.
Layers (BA-07)	Rows 56 dp: eye · lock · name · "100% · n objects" · row overflow (Rename, Export layer, Duplicate) · drag grip. Active layer: 4 dp blue bar + tint. Imported layer is read-only (lock shown, no delete of objects). Header overflow : Import layer.... Footer fixed 48 dp: + New layer · Delete layer (asks: move objects or delete them; undoable). Default layer cannot be deleted. Hint: top of list draws on top.
Symbols	Search field 40 dp (English/Arabic by name and tags), tabs Favourites · Recent · All, category chips scroll horizontally (15: General, People, Transport, Vehicles, Aircraft, Maritime, Logistics, Emergency, Buildings, Weather, Outdoor, Infrastructure, Range/Rings, Arrows, Callouts), 4-column grid of 64 dp tiles (SDF symbols, recolourable). No footer; a one-line hint sits at the bottom. Empty: "No symbols match".

States for every panel: loading (skeleton rows), empty (icon, one line, one action — "No layers yet — New layer"), error (inline message with Retry), disabled (controls dimmed, not hidden) in Presenter Mode where panels are closed.

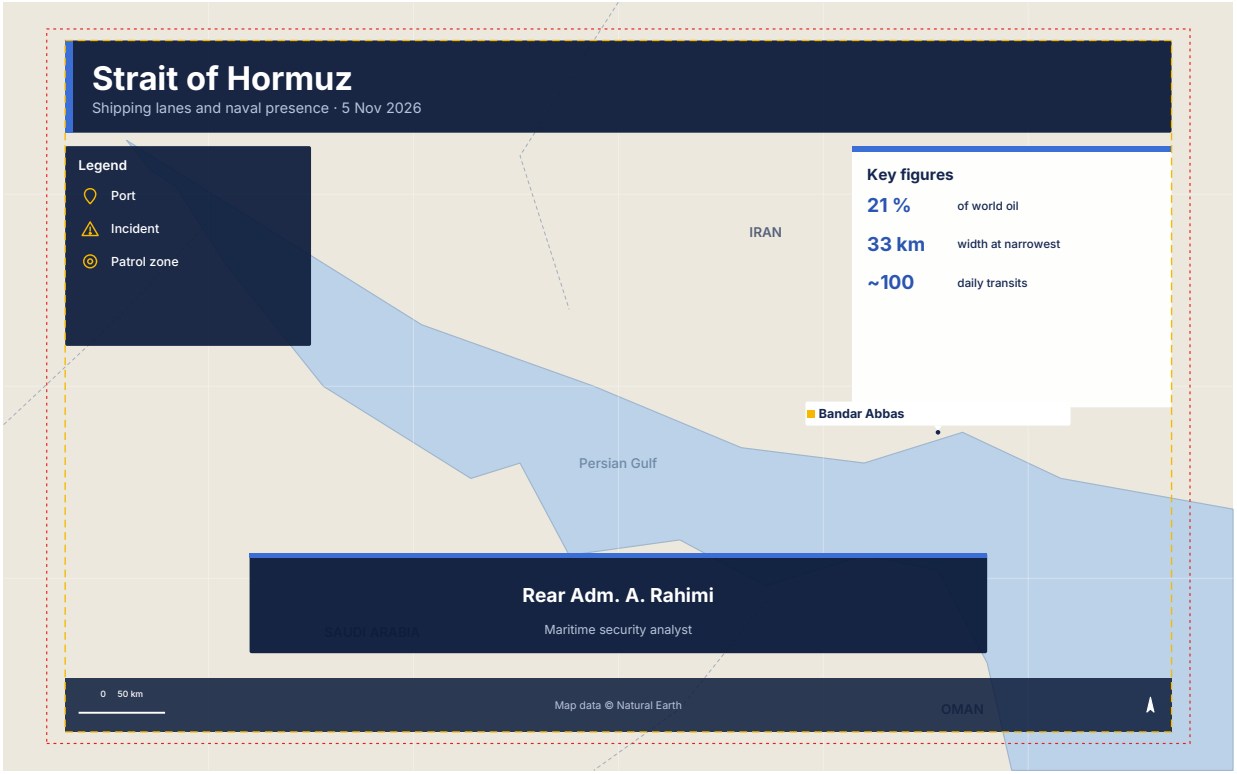
11.3a Scenes, Maps, GIS Import and Scene graphics: visuals



11.3 Scenes, Maps, GIS and Scene graphics: specification

Panel	Specification
Scenes (BA-13, BA-14)	Top card Presentation options: Remote control toggle (default off; server listens only while on) with Pair a phone… opening the QR dialog; Show my location toggle (default off). Scene rows 64 dp: thumbnail 72 × 48, number + name, “fly-to · n layers visible”, drag grip; live scene has a blue outline. + Add scene from the current view. Footer: Graphics · Present from here (primary). Operations: rename, reorder, update, delete, all undoable.
Maps — Map Sources (BA-08)	Tabs: Map Sources | GIS Import. + Import map source (primary) opens the format chooser: MBTiles · PMTiles; Image/scan/world file; Google Earth KML/KMZ; GeoTIFF · GeoPackage · ZIP tiles; each with a plain-language hint and “Where do I find this file?” (other apps' storage is private: copy to Downloads first). Rows 64 dp: thumbnail, name, format · size · “in use”; conversion rows show a determinate bar and Cancel (never a blocking spinner). Details card: bounds, zoom, CRS, SHA-256, attribution; actions Rename, Delete, Export source, Georeference. Deleting a source in use is blocked or warned.
Maps — GIS Import	Choose file (GeoJSON, KML, KMZ, GPX, CSV) → detected format and CRS, feature counts, mini-map preview, warnings (“3 features skipped (invalid geometry)”), “Imported layer is read-only”. Footer: Cancel · Import layer. CSV shows a column-mapping step for lat/lon. Convert to MapX objects is a later, undoable action from the context bar.
Scene graphics	Per-scene toggles and text for the seven graphics (§11.4), Transition (Instant · Fly-to · Fade · Appear), and Preview exactly as audience sees it in the footer. Panel closes when presenting.

11.4 Presentation graphics: seven graphics in a news-channel style



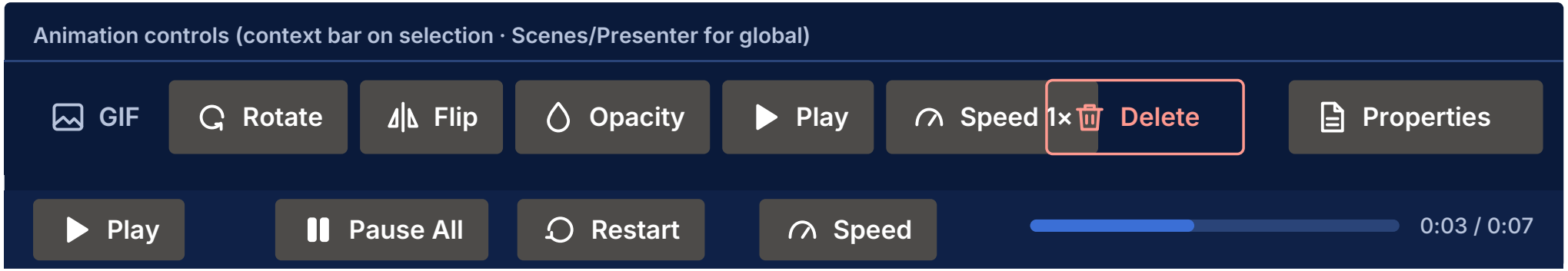
Zones with title-safe (5%, amber) and action-safe (3.5%, red) margins

Zone	Graphic	Max size
Z1	Headline banner	14% h, full title-safe width
Z2	Legend	22% w × 35% h, inline-start
Z3	Information card	28% w × 40% h, inline-end
Z4	Lower third	60% w × 16% h, centred, above the strip
Z5	Scale · Source line · North	Strip 9% h, in this order from inline-start
—	Location label	Anchored to its coordinate; drawn below screen graphics

- Style: restrained broadcast look: solid navy plates at 92–95% (white plate for the info card), one 5–8 dp blue accent bar, Cairo Bold headlines, 2 dp radius. No shadows, gradients or animation beyond the chosen transition.
- Readable on any map: every graphic sits on an opaque or 92%+ plate; text ≥ 4.5:1 on its plate (white on navy 17:1; navy on white 16:1).
- Hierarchy: headline 34 sp → lower-third name 20 sp → body 13–15 sp → captions 11 sp (at 1280 px width, scaled by width).
- Collisions impossible: text truncates inside its zone; zones never grow; Z4 starts below Z2/Z3. The scene editor warns if a location label falls under a zone.

12 Images and GIF overlay interface

Images and animated images are geographically anchored map objects: centre + width in metres + rotation. They scale with the map and are edited with the same handles as any object.



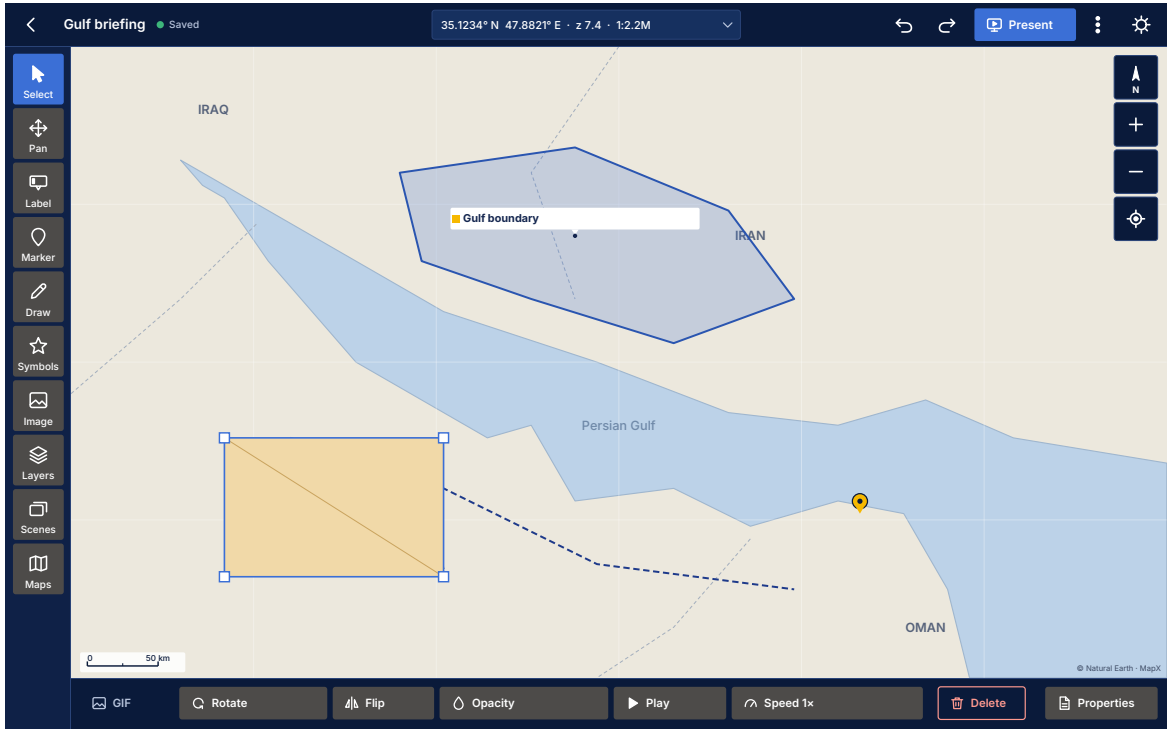


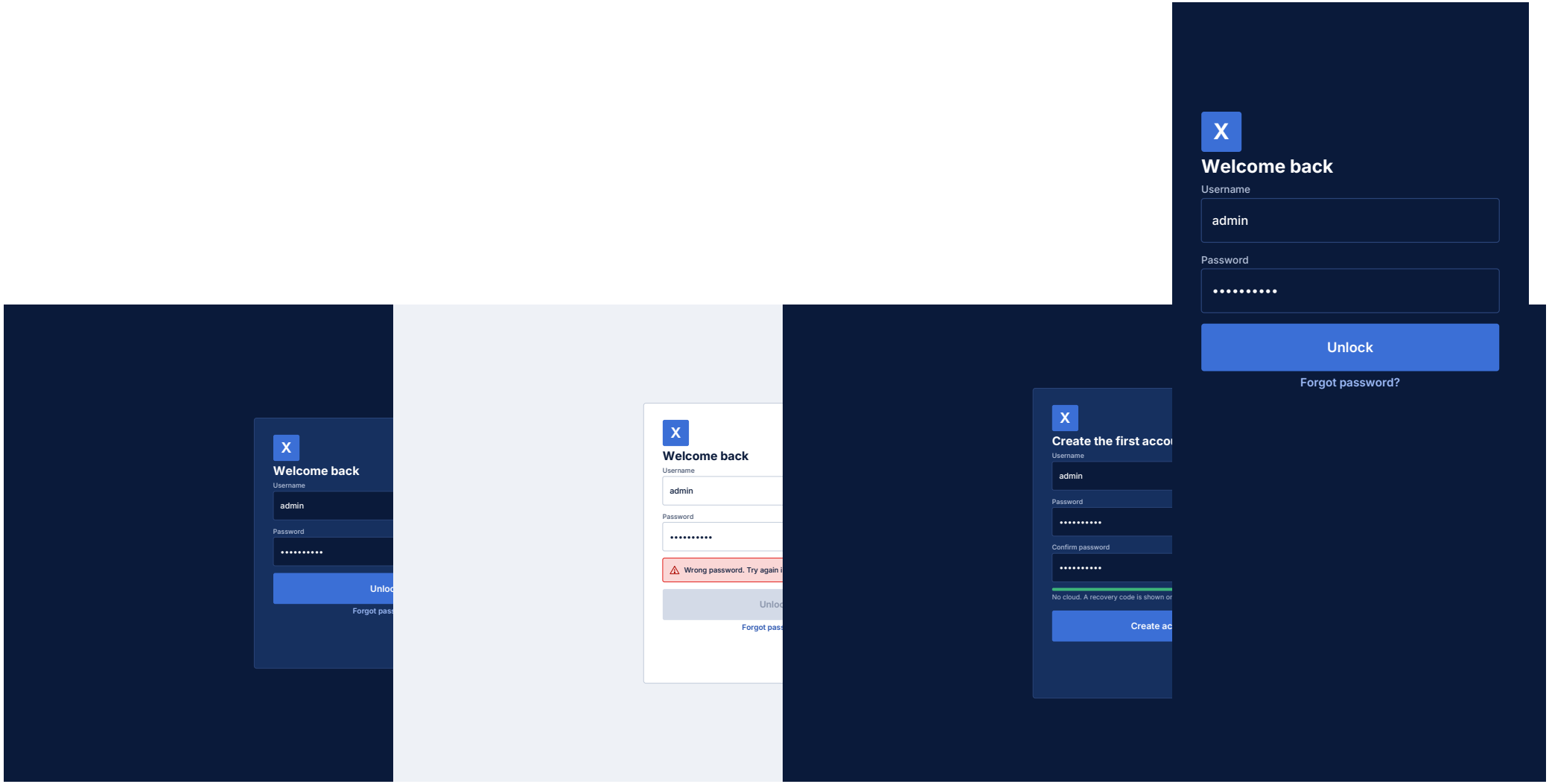
Image selected: 4 corner handles, 48 dp hit areas; GIF actions in the context bar

Flow / state	Specification
Import	Image tool → system file picker (PNG, JPG, WebP, GIF, animated WebP). Placed at the map centre, ~25% of map width, selected with handles. Duplicate media (hash) are reused.
Edit	Move, scale, rotate, flip, opacity, lock, delete; one gesture = one undo step. Large handles on tablet; on phone a contextual sheet.
GIF / animated WebP	Context bar adds Play/Pause and Speed (0.5× · 1× · 2×); Properties adds Loop and a progress scrubber. Global: Play, Pause All, Restart, Speed.
Visible-only playback	Off-screen, hidden or hidden-layer animations neither decode nor advance; paused on background. Cap of 6 concurrent (configurable): the 7th shows a paused badge and a snackbar “Animation limit (6) reached”.
Reduced motion	System reduced-motion pauses all animation on the first frame; Play is still available manually.
Missing media	Placeholder tile with a warning glyph and “Image missing — Relink / Delete”; never a crash.
Large photo	Decoded with downsampling (≤ 2048 px display); shows a thin progress bar while decoding.
KMZ GroundOverlay	Imported as an image object at its geographic box and rotation.

13 Settings and other required screens

Every screen and flow in the specification that is not the editor: Lock, Settings, Accounts, Georeference, Export/Import, Presenter, Audience, Remote and Pairing.

13.1 S0 Lock / Login (BA-02)

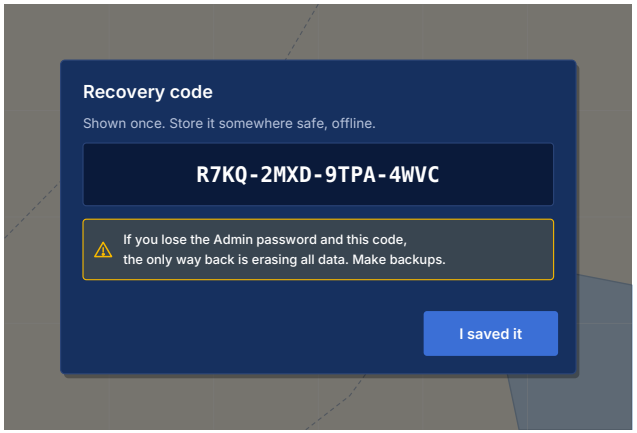


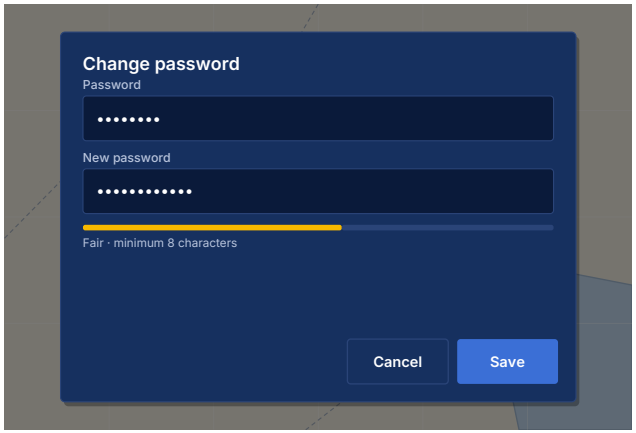
Tablet · sign in | Lockout (light theme) | First run: create Admin | Phone

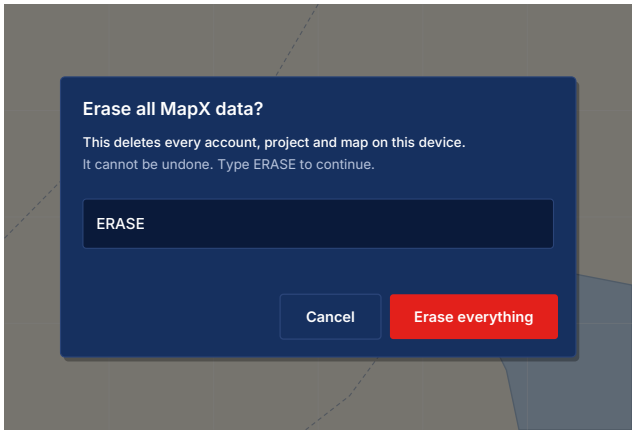
Aspect	Specification
Layout	Tablet: centred 440 dp card on the navy background. Phone: full width, 16 dp margins, content top-aligned at 120 dp. No rail, dock, panels, context bar; nothing from projects is visible. Window is excluded from screenshots and recents.
Fields / actions	Username, Password (48 dp fields, show/hide), primary Unlock 52 dp; “Forgot password?” (Member → ask Admin; Admin → recovery code). Biometric button appears only after enabling it (P2).

Aspect	Specification
Lockout (FR-SEC-05)	After 5 failures: 30 s, doubling to 15 min, persisted across restarts. The button becomes disabled with a live countdown and a red-outline notice; no data wipe.
First run (FR-SEC-02/07)	Create the Admin: username, password ×2, strength bar (min 8, no composition rules), then a one-time Recovery code dialog. If no device screen lock is set, a warning card explains that login alone cannot protect storage.

13.1 Account dialogs (BA-02, S10)

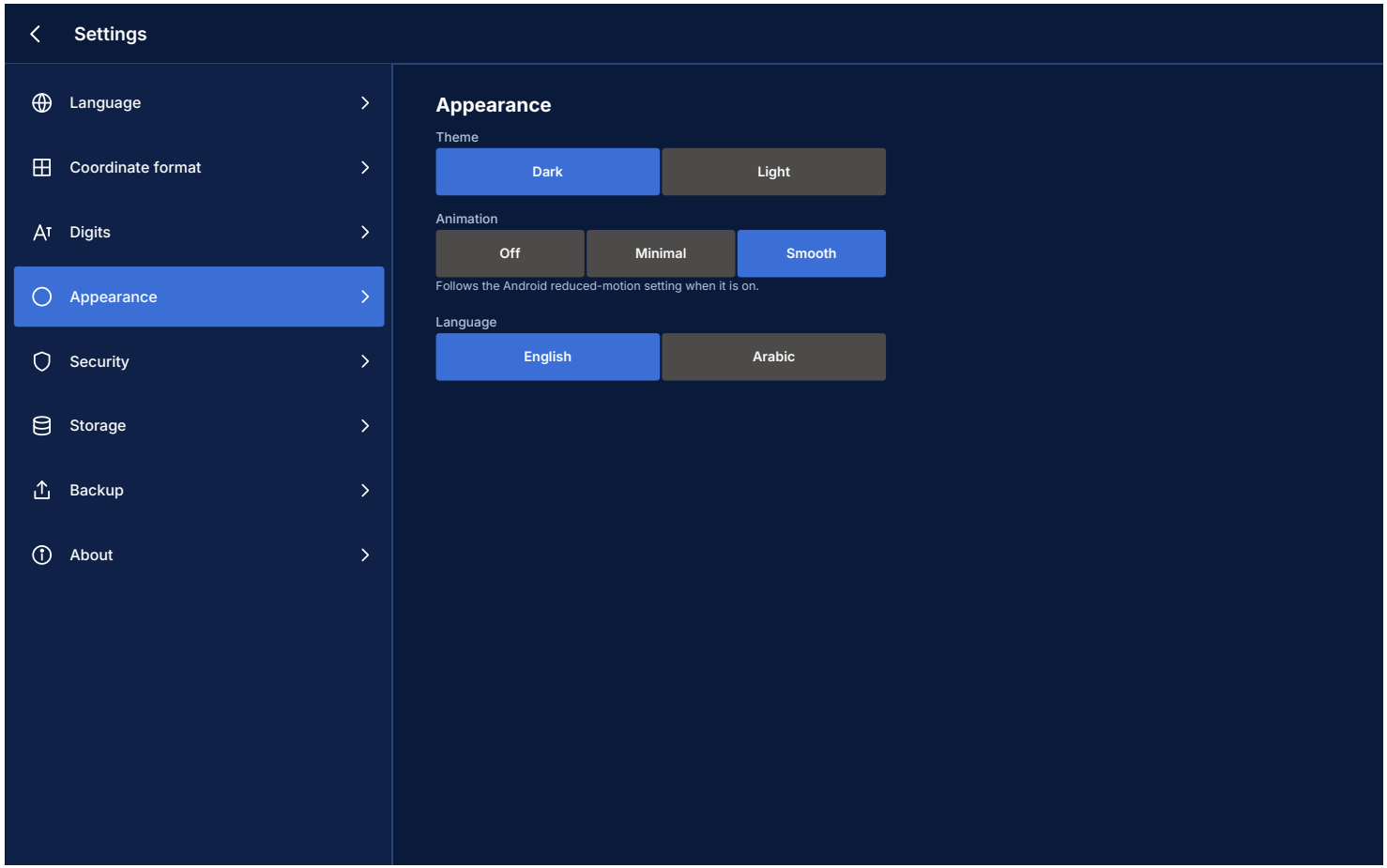




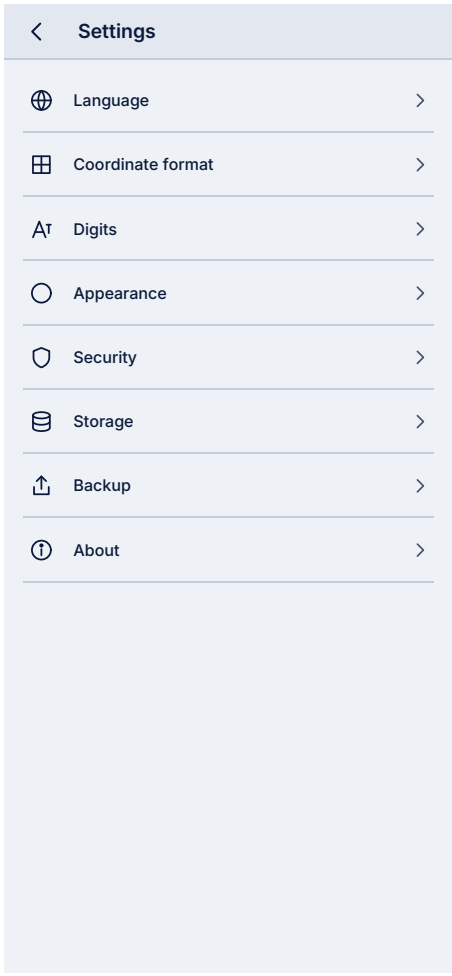


Dialog	Rules
Create account / Recovery code	Username + password; Admin gets a one-time code (monospace, 19 sp, copy-less by design). The dialog states that a lost password plus code means starting over.
Change password	Current + new + strength; saved with a snackbar; ends other sessions on this device.
Erase all data	Heavily confirmed: states consequences, requires typing ERASE; destructive button is red fill, Cancel first.
Accounts (S10, Settings → Security)	List of accounts with role chips (Admin/Member); Admin can add, rename, reset a Member's password, delete (confirmed: deletes that account's data). Each account has its own project and map libraries.

13.2 S7 Settings (BA-12)



Tablet: 360 dp list + detail pane

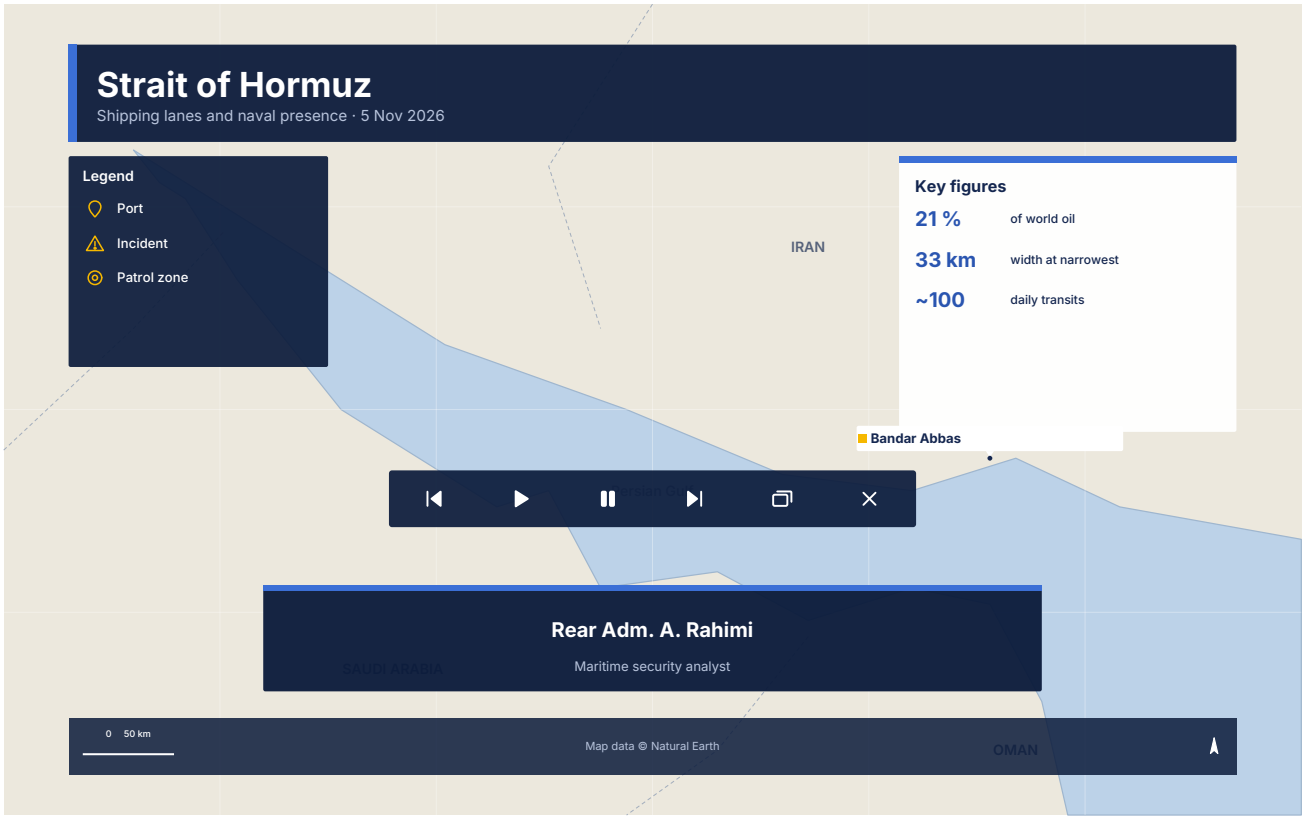


Phone: single column

Section	Content (FR-SET-02 + owner decisions)
Language	English (LTR) / (RTL); applies immediately, persists (DataStore).
Coordinate format	DD · DDM · DMS · UTM · MGRS; MGRS precision 10 km … 1 m (default 1 m). Metric units are fixed (no unit switch).
Digits (P2)	Western (default) / Arabic-Indic display.
Appearance (DEV-5)	Theme Dark / Light; Animation Off / Minimal / Smooth (system reduced-motion overrides).
Security	Auto-lock (1, 5, 15, 30 min, Never with warning; never during presentation), Protect screen, Change password, Accounts, optional biometrics (P2), Lock now.

Section	Content (FR-SET-02 + owner decisions)
Storage	Usage per project and per map source; cleanup of unused media and orphan files (confirmed).
Backup	Backup all (.mapxbackup: optional sources, optional passphrase), Restore (add as new / replace, confirmed), backup reminder (P2).
About	Version, licences (Cairo OFL, Natural Earth, MapLibre BSD), symbol licences. Debug builds add Offline-guard status.

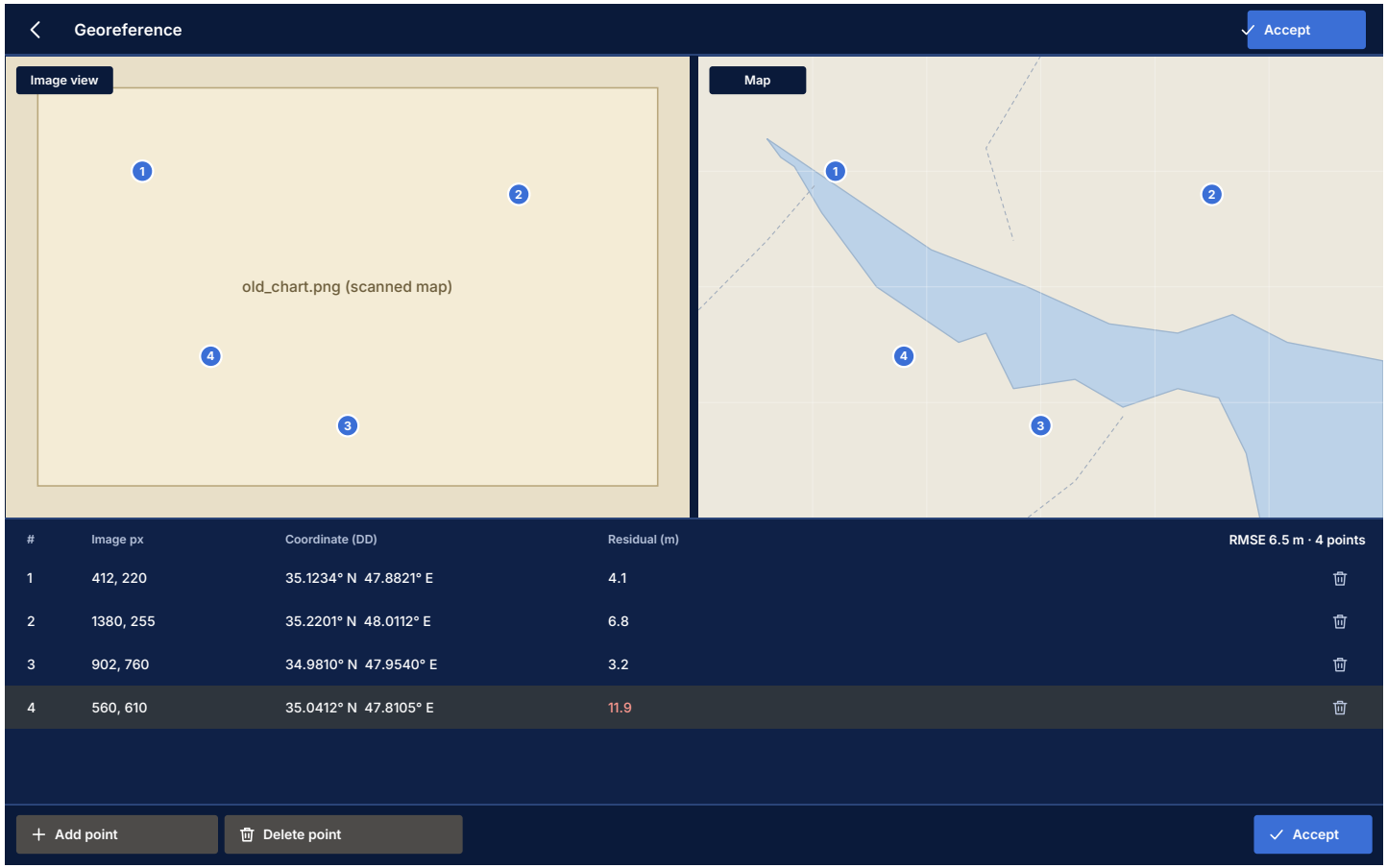
13.3 S4 Presenter Mode and transient controls



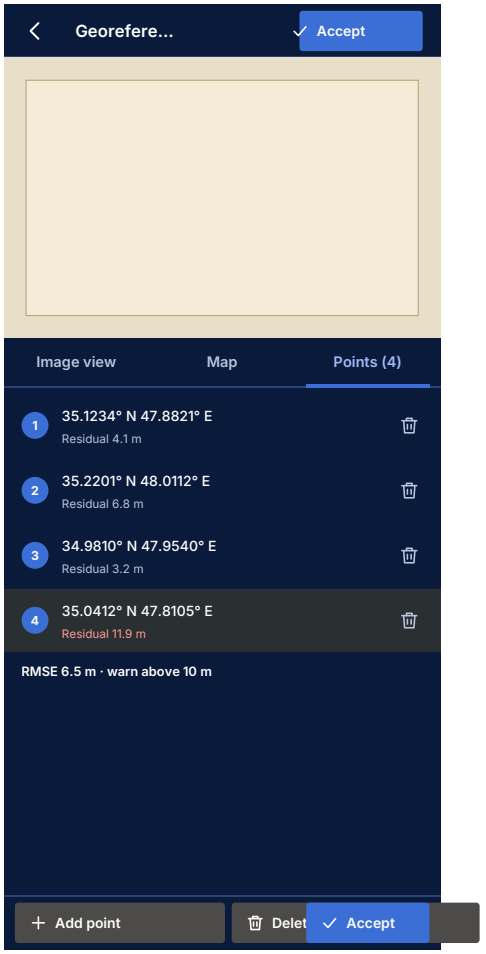
Presenter Mode (tablet): transient controls bar (tap to show, hides after 4 s)

- No rail, panels, handles, selection or context bar. Immersive bars, screen on, rotation handled.
- Transient bar 56 dp, 520 dp wide: Previous · Play · Pause · Next · Scene list · Exit. Exit needs a deliberate action; Back asks to confirm.
- Graphics occupy zones Z1–Z5 (§11.4); location labels are geographic, drawn below them.
- Location dot hidden unless Show my location is on.
- Presenter/remote camera and layer overrides are session-only: never saved.
- Editor preview shows the same render with Prev · Next · Play · Pause · Exit preview.

13.4 S8 Georeference (BA-09, FR-GRF-01..06, BA-16)



Tablet: image | map views synchronised, point table below

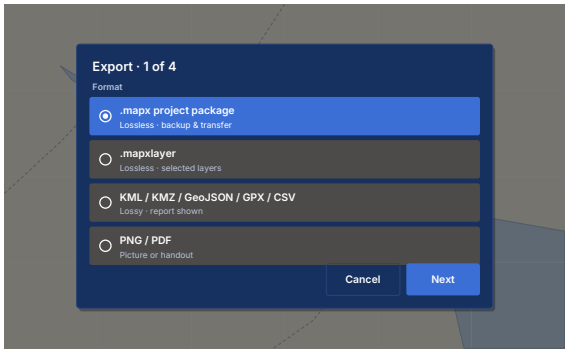


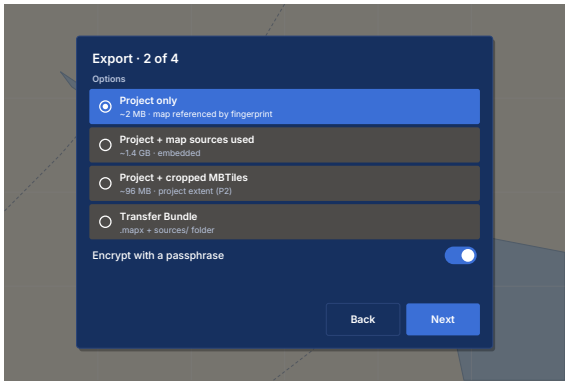
Phone: tabs

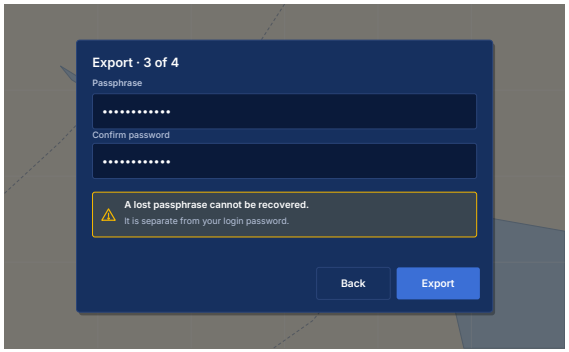
Aspect	Specification
Regions	Top bar 48 (Back · title · Accept) · two equal views (each ≥ 45% of the width) · table (header + 40 dp rows, scrolls) · context bar 56. Siblings only; pinch/pan in one view mirrors the other's extent; the lock icon between views toggles sync.
Control points	≥ 3 non-collinear points, numbered badges on both views. Add point: tap image, then tap map (or type coordinates in any format). Select a row to highlight both markers. Drag to adjust; Delete point removes the selected row.
Residuals	With ≥ 4 points each row shows its residual in metres; the worst row is tinted amber; the footer shows RMSE and a warning for large RMSE or poor distribution.

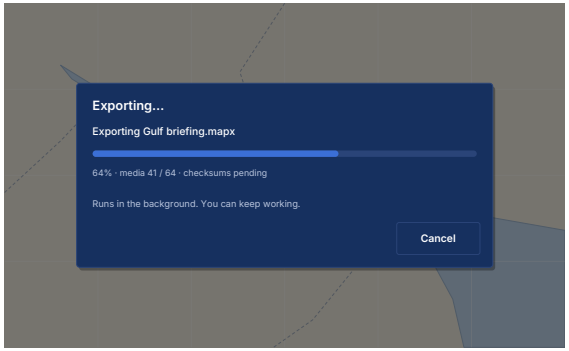
Aspect	Specification
Decisions	Accept stores transform, points, RMSE and CRS with the source; Adjust returns to editing; Reject discards. Converting IMAGE_LOCAL objects to geographic is a separate, undoable action.
Phone	Tabs: Image · Map · Points (n). Top view area 250 dp; point rows 56 dp with residual line; context bar 48 with Add point · Delete point · Accept.

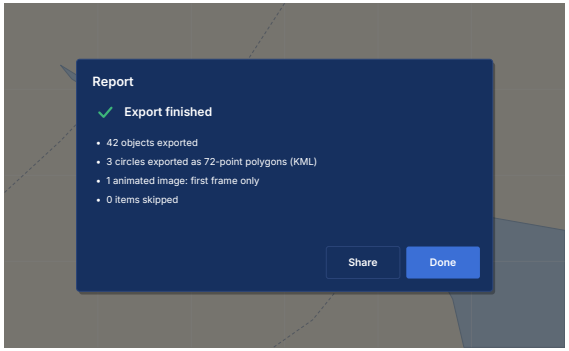
13.5 S9 Export wizard, passphrase, progress, report and relink (BA-10)

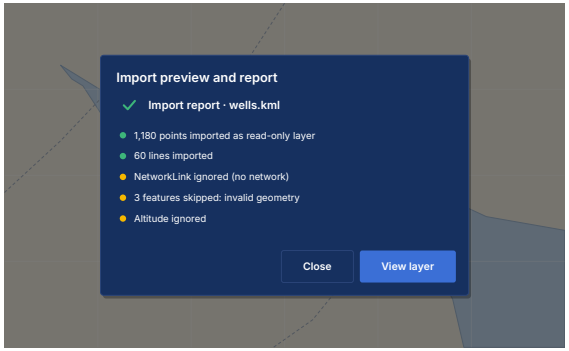






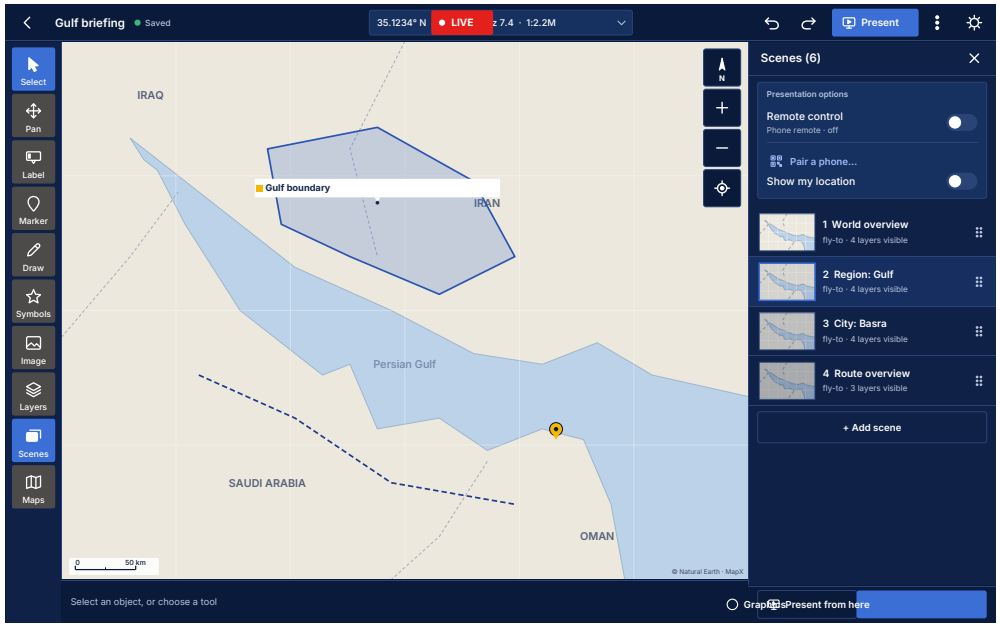


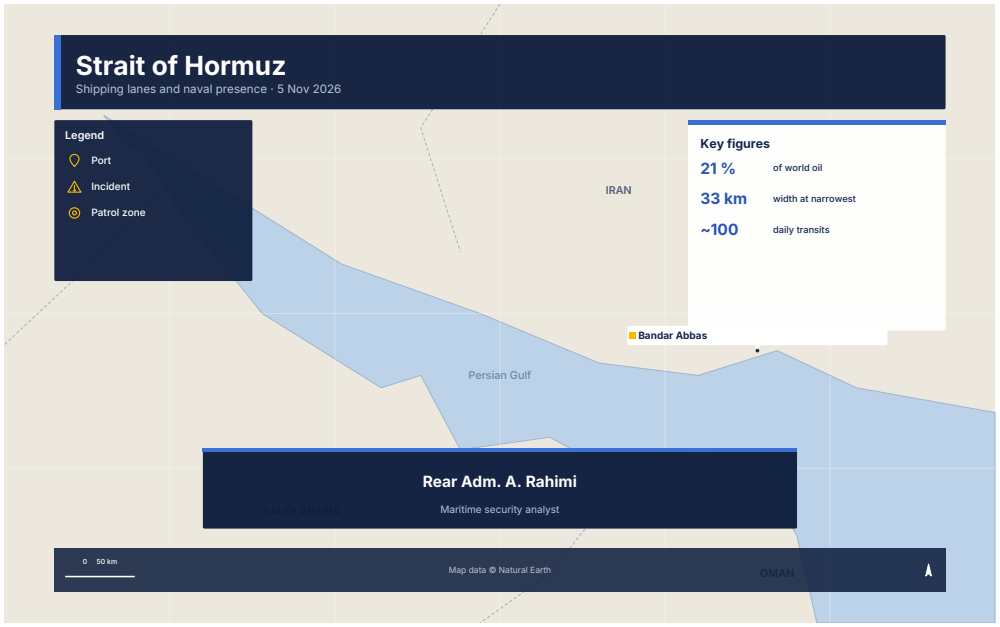




Aspect	Specification (problem H)
One dialog, four steps	Format (.mapx, .mapxlayer, KML/KMZ/GeoJSON/GPX/CSV, PNG/PDF) → Options (Project only · Project + map sources · Project + cropped MBTiles (P2) · Transfer Bundle, each with a size estimate; Encrypt toggle; visible-only layers) → Passphrase (only when encrypting; "A lost passphrase cannot be recovered") → Progress → Report. Steps scroll inside the dialog; the dialog is 520 dp on tablets and a full-height sheet (width – 0, top inset 24) on phones.
Progress and results	Determinate bar with Cancel; runs in the background; the editor stays usable. Finish shows the report (counts, lossy conversions, skipped items) and a snackbar "Export finished [Share]". Nothing is silently dropped.
Import preview / report	Opened by open-with, Import..., Home → Import project package...: verifies checksums and version first, shows what will happen (new copy with new ids if the id exists), then a report. Newer format: "Update MapX". Wrong passphrase is detected before extraction.
Relink map source	Opens when a project's source is missing (name, format, size, bounds shown): choose from library or import a file with the matching fingerprint, or open without a map. Objects never move.

13.6 S5 Audience screen and external display (BA-04)

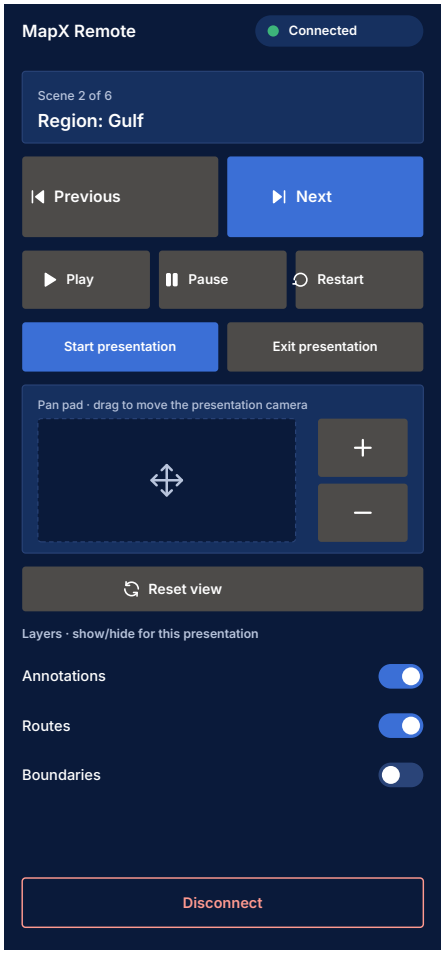




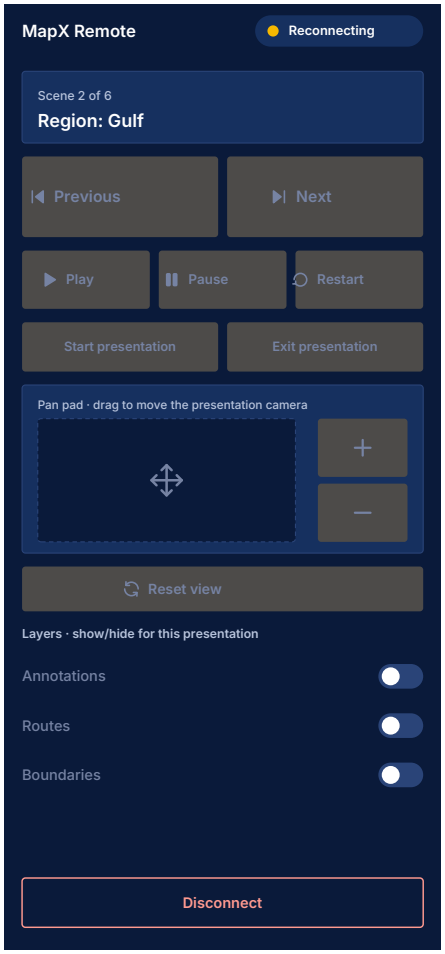
Tablet = editor and controller (LIVE chip, Scenes panel). Right: audience display = clean output only, its own MapView.

Case (FR-DSP-06)	Interface behaviour
Live	Top bar shows a red LIVE chip (80 × 32) and Stop; Scenes panel marks the live scene and the next scene preview. The tablet stays fully interactive.
Audience content	Map, annotations, media, animations and graphics only. Never the rail, panels, handles, selection outlines, hidden layers or editor-only UI state (BA-04).
Cable removed	Snackbar “Display disconnected” with Reconnect and Use in-app preview; the project is untouched.
No display / mirroring only	Honest message; a draggable in-app Audience Preview window (same clean render) is offered. Mirroring-only is stated as such.
Resolution change	Zones re-layout by percentage; nothing is clipped. The audience window is never screenshot-protected (it would turn black).

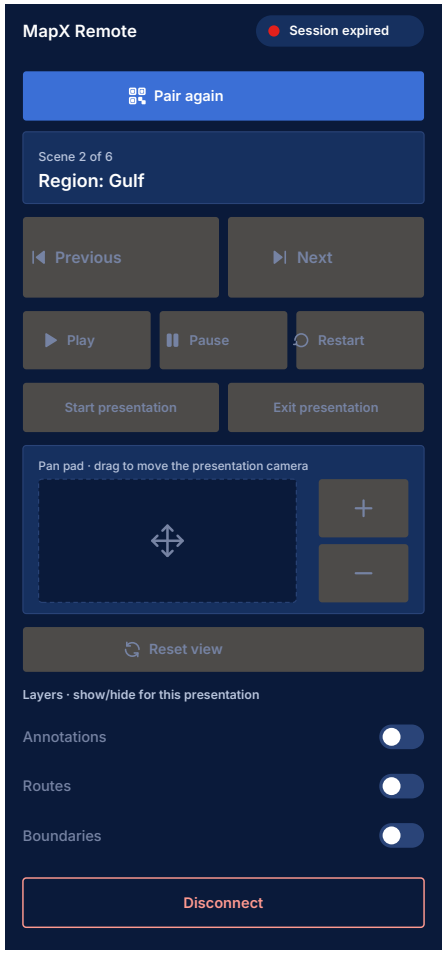
13.7 S6 Phone remote controller (BA-03)



Connected



Reconnecting (disabled, not hidden)



Session expired: Pair again

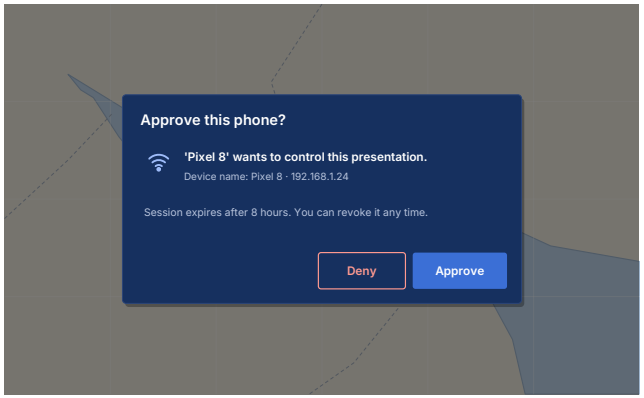


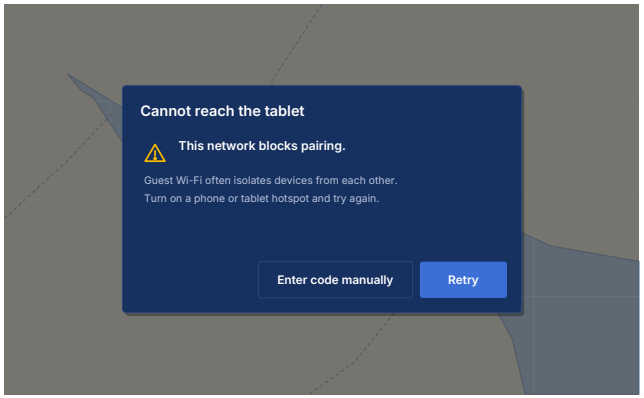
Light theme, Arabic RTL

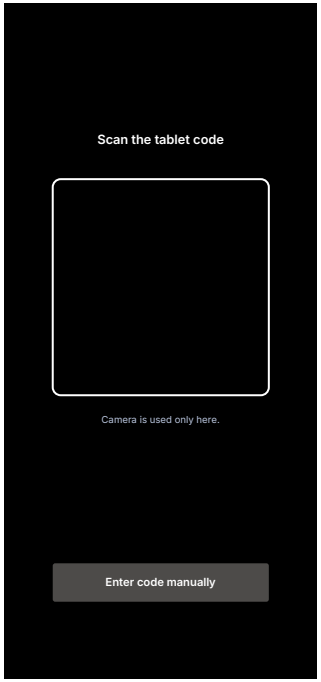
Layout. One scrolling column: status chip → current scene card → large Previous/Next (72 dp) → Play/Pause/Restart (52 dp) → Start/Exit presentation → pan pad with zoom → Reset view → layer toggles labelled “show/hide for this presentation” → scene list → Disconnect. Removed from v1: Undo, Redo, Tool. In landscape: two columns (scene and playback at the start edge; pan pad, zoom and toggles at the end edge). In every state except Connected all controls are disabled, never hidden. Status chip: Connected (green) · Reconnecting (amber) · Disconnected (grey) · Session expired (red + Pair again). The phone has its own login first; no drawing, editing, rail or panels.

13.8 S11 Pairing flows (BA-17, BA-14)









Phone: QR scanner

Dialog	Content
Pair a phone (tablet, from Scenes → Remote control)	QR (host, port, certificate fingerprint, one-time secret), 4 numbered steps, manual code with 2-minute expiry, Cancel. Remote control must be switched on first.
Approve this phone	Shows the device name and address (“Pixel 8”), session length (8 h) and revocability; Deny · Approve. A second phone requires explicit takeover.
Cannot reach the tablet	Network blocks pairing help: guest Wi-Fi isolates devices; use a hotspot. Buttons: Enter code manually · Retry.
Scanner (phone)	Camera permission asked only here with a one-line reason; manual code entry is always available; locking either device ends the session.

14 Responsive layouts

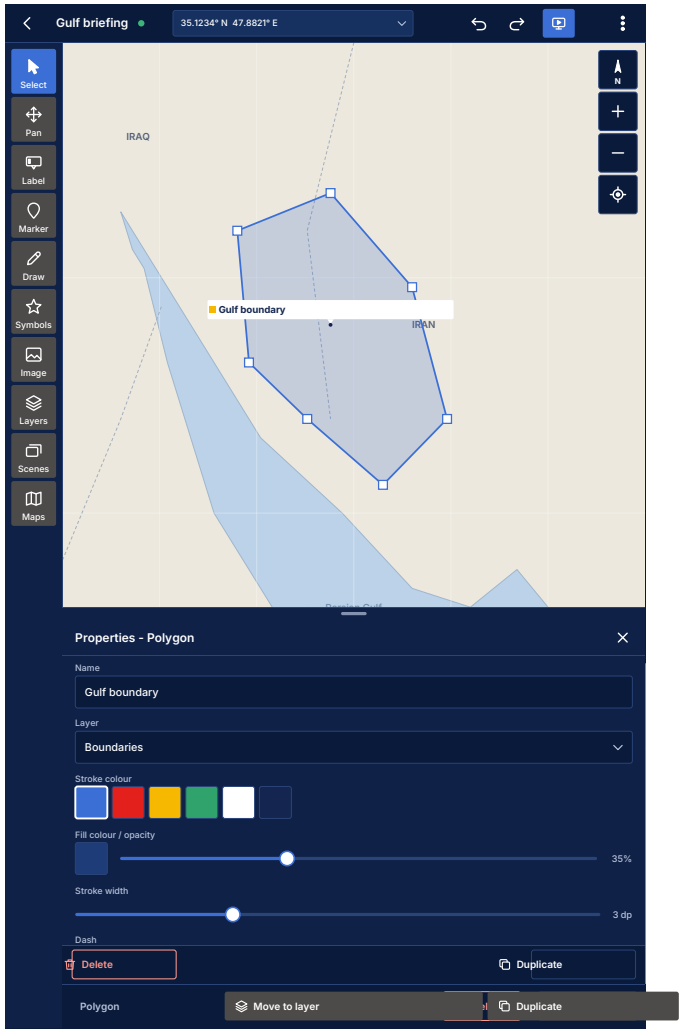
One LayoutSpec drives every dp value. Phones are designed as phones; tablet landscape is the primary target.

14.1 LayoutSpec v1.1: size table (changes from Blueprint v1 in bold)

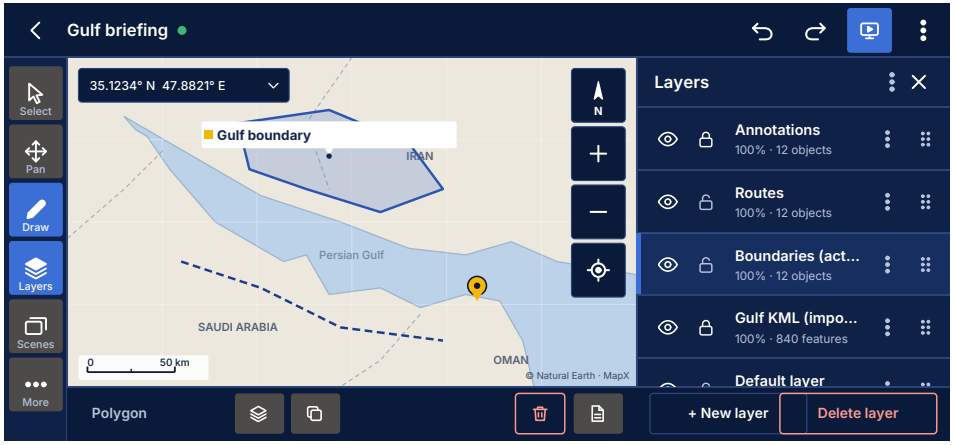
Region	TABLET_LANDSCAPE 1280×800	TABLET_PORTRAIT 800×1280	PHONE_LANDSCAPE 844×390	PHONE_PORTRAIT 390×844
TopStatusBar	48	48	48 (was 40)	48 (was 56)
ToolRail	72 wide · 10 tiles 56 · gap 4	72 wide · 10 tiles 56	56 wide · 5 slots + More · items 48 (was 44)	none
BottomDock	none	none	none	56 (was 64) · 5 slots: Select, Pan, Draw, Layers, More
BottomContextBar	56 under the map	56 under the map	48 (was 44) under the map	48 above the dock, only while a tool/selection is active
SidePanel	320 fixed (min 280, max 360) + close	none (sheet)	280	none (sheet)
BottomSheet	none	peek 80 · half 40% · full	none	peek 72 · half 50% · full
Map share	≈ 82% (60% with panel)	≈ 83% (≥ 50% with half sheet)	≈ 75% (≥ 45% with panel)	≈ 83% idle, 76% active (≥ 35% with half sheet)
Touch targets	≥ 48 (tiles 56)	≥ 48	≥ 48	≥ 48

Phone portrait budget (dp)	360×640 window	412×915 window
v1: top 56 + dock 64 (+ context 48)	map 520 idle · 472 with context bar	map 795 idle · 747 active
v1.1: top 48 + dock 56 (+ context 48)	map 536 idle · 488 active (+64 / +16 dp)	map 811 idle · 763 active
Same minus system bars (≈ 24 status + 24 gesture)	map 488 idle · 440 active	map 763 idle · 715 active
Half sheet (50%)	visible map 244 idle / 220 active: controls compact to compass + Locate when < 232	visible map 405 / 381: full 4-tile stack

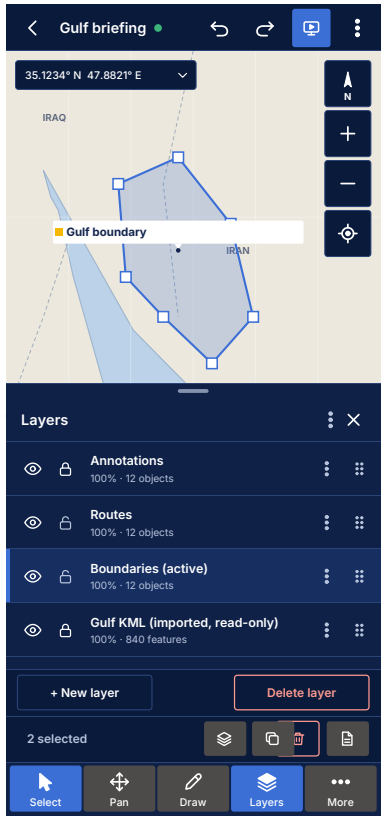
14.2 The four modes



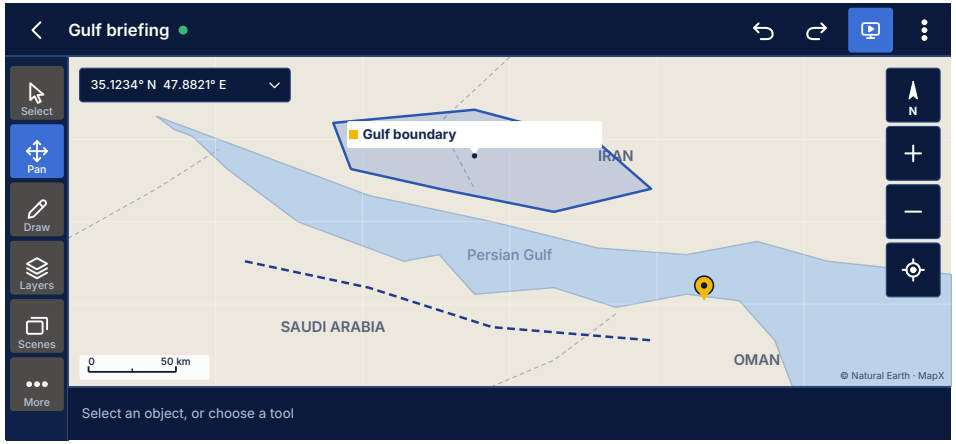
Tablet portrait + half sheet (40%)



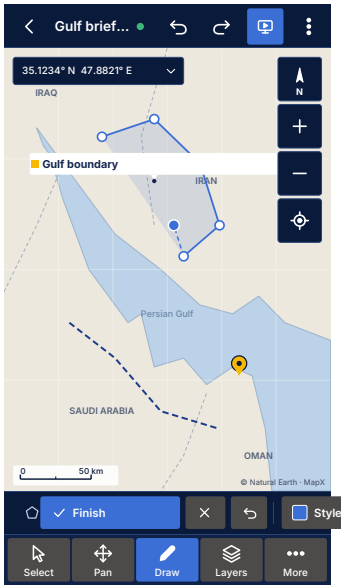
Phone landscape + 280 dp panel

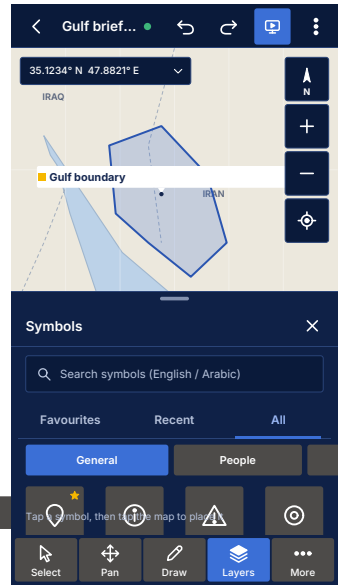


Phone portrait + half sheet



Phone landscape: rail 56 · top 48 · context 48

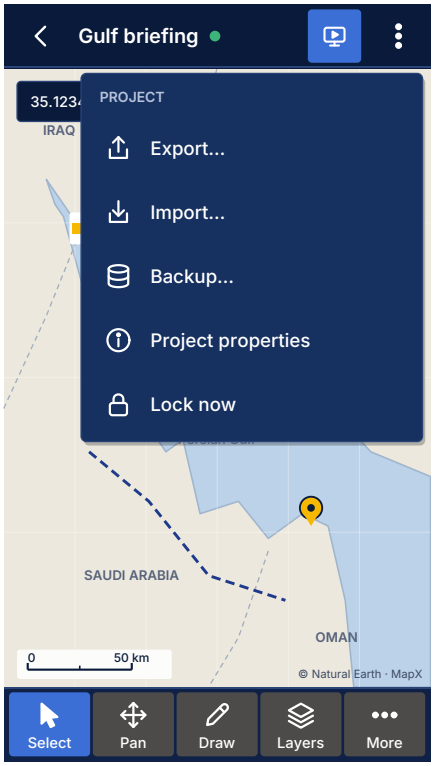




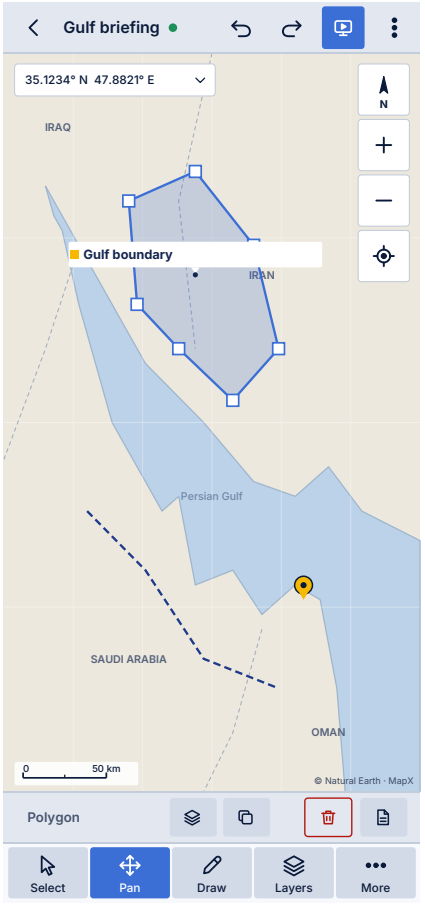
360 × 640: drawing; context bar appears, map shrinks, bottom sheet

360 × 640, half-sheet controls compact

14.3 Narrow phones, stress sizes, font scale and split-screen



320 dp wide: Undo/Redo in the Project menu



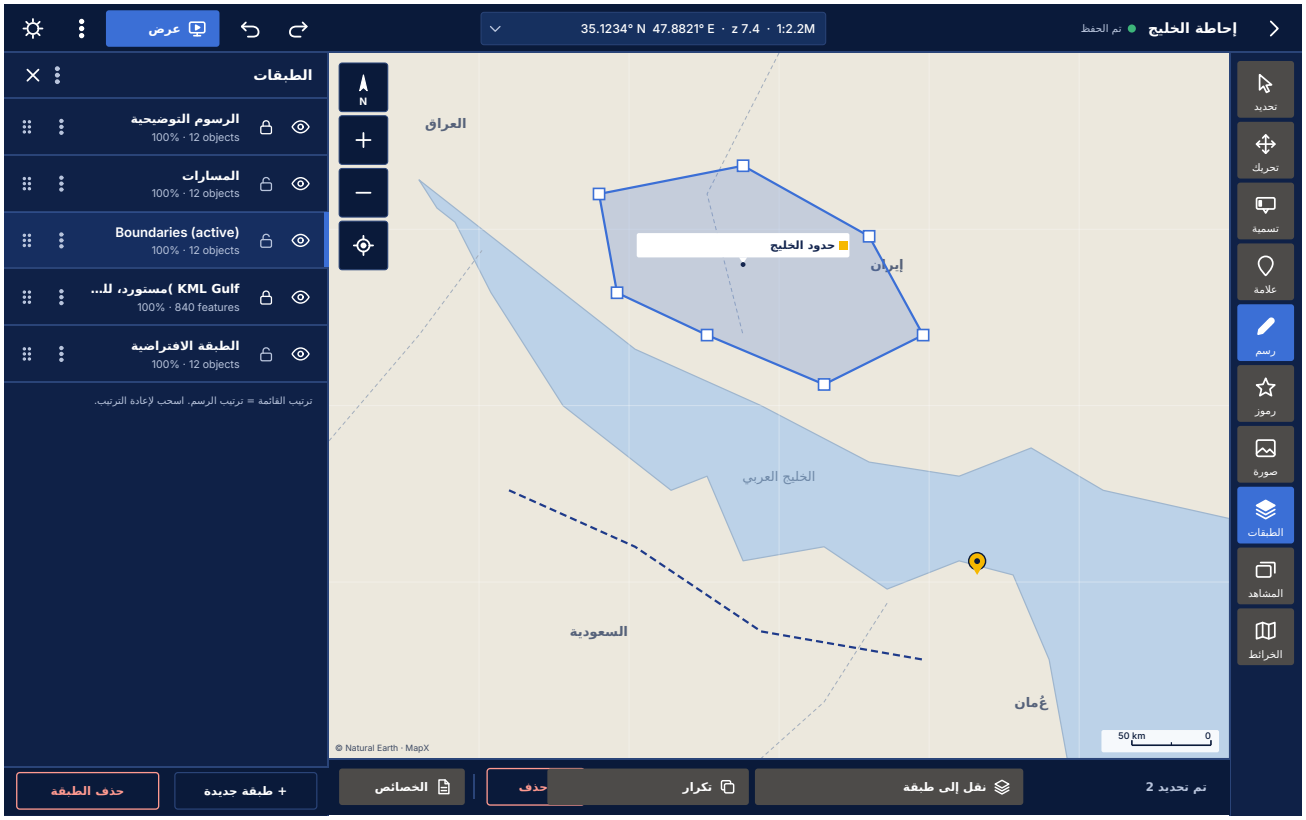
Phone portrait, light theme

Stress size (dp)	Mode	Notes
1280×800 · 1600×1000 · 1920×1200	TABLET_LANDSCAPE	Panel stays 320; map takes the rest.
800×1280 · 1200×1920	TABLET_PORTRAIT	Panels as bottom sheet; rail stays.
360×640 · 390×844 · 412×915	PHONE_PORTRAIT	Top 48 / dock 56; 320 dp adds the narrow top bar.
844×390 · 915×412	PHONE_LANDSCAPE	Rail 56 / side panel 280; keep out of the cutout side.
Font scale 200%	any	Bars keep height and truncate/scroll; bodies wrap and scroll; panel width fixed; no second toolbar row.
Split-screen / resize	any	Re-classified on the window; a tablet below 600 dp is PHONE; panel ↔ sheet keeps scroll and field input.
Cutouts / insets	any	Insets consumed once at the root; rail/dock stay clear of the cutout; map controls stay in the safe area.

Compose contract (unchanged from Blueprint §14): one root Column [TopStatusBar][Body]; Body is a Row (rail, map column, side panel) or a Column (map box with sheet, context bar, dock); ModalLayer is the last child. `LayoutSpec.of(mode)` holds every dp value above; no screen declares its own. Add the `regionsNeverOverlap` test for every mode in both languages and add modes PN/PS (360, 320 dp) to the stress list.

15 RTL and localization specifications

Arabic is a full mirror of the layout, not a translation of labels. All geometry is start/end; the map, compass and media-play icons never flip.



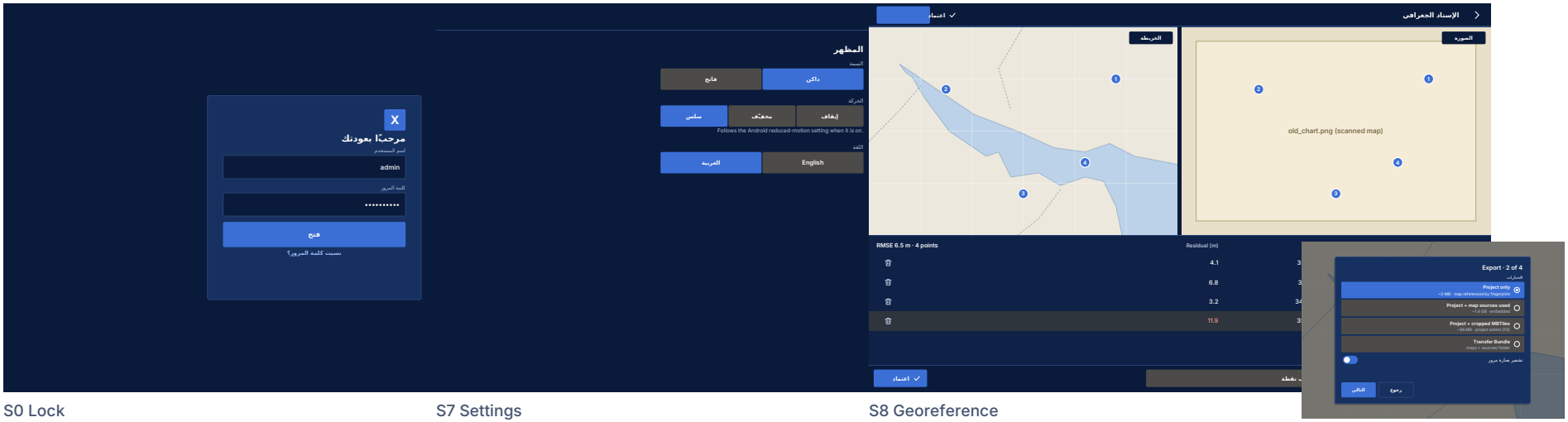
Arabic RTL, tablet landscape (dark): rail at the right, panel at the left, controls at the end (left) edge



Arabic RTL, phone (light)

Blueprint correction (DEV-7). Blueprint Fig. 15 says map controls “move to the start (right) edge”. By Law L10 they stay at the logical end edge: right in LTR, left in RTL, next to the side panel. The scale bar and attribution swap the other way.

15.1 Mirroring rules, new screens (BA-15) and Arabic text



S0 Lock S7 Settings S8 Georeference

Mirror	Flip	Never flip
Rail ↔ panel sides, top-bar order, map-control and scale-bar sides, context-bar order, lower-third and legend zones (Z2/Z3 swap, Z5 reverses), sheet handle alignment, table column order, dialog button order (primary stays at the end edge)	Back, Next/Previous, Undo/Redo, skip icons, list chevrons, progress and slider fill direction, import/export arrows	Map and compass, play/pause, clocks, numbers (coordinates, scale, counts), Latin brand names, N/E/S/W letters, QR codes

- S0: card centred; fields right-aligned; show/hide icon at the start (right). S7: list on the right, detail on the left; segmented controls reverse order. S8: image view at the start (right), map at the end; table columns reverse; residual numbers stay LTR. S9: steps run right-to-left; Back at the start, Next at the end.
- Bidi: mixed Arabic/Latin/number text uses bidi-safe rendering; coordinates are LRI-isolated; the Label Tag anchor square sits on the inline-start (right) edge; 40% text expansion must not clip.
- Verify with the existing RTL checklist (LC-2) plus pseudo-locales en-XA and ar-XB at all ten stress sizes plus 360 and 320 dp.

English	(illustrative)	English	(illustrative)	English	(illustrative)
Select	تحديد	Pan	تحريك	Draw	رسم
Layers	الطبقات	More	المزيد	Marker	علامة
Label	تسمية	Symbols	رموز	Image	صورة
Scenes	المشاهد	Maps	الخرائط	Present	عرض
New Project	مشروع جديد	Open Project	فتح مشروع	Settings	الإعدادات

A Validation record

Checked against the brief's Stage 5 list. “Verified” means checked in this document or by the automated region check; nothing was run on a device.

#	Check	Result	Evidence
1	All required screens covered	Verified	S0–S11 in §2.1; layouts in §8, §9, §13; Presenter/Audience/Remote/Pairing in §13.3, 13.6–13.8.
2	All 17 Blueprint amendments	Verified	BA-01 … BA-17 register on page 5 with section references.
3	All editor tools	Verified	Ten rail tools §11.1; eleven drawing tools §10.1.
4	All seven panels	Verified	Properties, Layers, Symbols, Scenes, Map Sources, GIS Import, Scene graphics (§11.2–11.3).
5	All seven broadcast graphics	Verified	Headline banner, Lower third, Location label, Legend, Scale + North, Source line, Info card (§11.4). The spec lists scale and north separately; they share one toggle here.
6	Phone and tablet layouts	Verified	§14: four modes plus 360 and 320 dp.
7	Dark and light themes	Verified	§3 tokens; both themes mocked in §8, §13, §14, §15.
8	Arabic RTL	Verified (design)	§15 plus RTL mock-ups of editor, lock, settings, georeference, export. Not tested with real Cairo text or TalkBack.
9	Colour accessibility	Verified (computed)	§3.1; fills with white text pass at 4.68–4.74:1 with thin margin; text tints ≥ 4.9:1.
10	Interactions and states	Verified	Locate (6 states), connection (4), save (4), lock-out, empty/loading/error rules in §9–§13.
11	No elements overlap	Verified	Automated region check on the guide's own layout model: 888 pairwise/containment checks over 6 modes × LTR/RTL × 4 panel states × 2 context states; failures: 0. Mock-ups were also inspected visually. The Compose test (LC-10) remains to be written and run.
12	No essential functionality missing	Partly	All FR-* UI surfaces are covered. Not designed (not UI in the spec): biometric unlock dialog (P2), direct nearby transfer (P2), TalkBack list view (P2), stacked map-source opacity UI (P2), CSV column mapping and MGRS precision pickers are described but not drawn.
13	Readable, professional PDF	Verified	Vector artwork, selectable text, page numbers, contents and PDF bookmarks.

Decisions that need the owner

ID	Question	Recommendation
DEV-1/2/3	Approve the reduced bar heights (phone portrait top 48, dock 56; phone landscape top 48, context 48)?	Approve: +16–64 dp of map and spec-consistent 48 dp targets.
DEV-4	Phone landscape rail: Symbols in More, Pan and Scenes visible?	Approve.
DEV-5	Add Appearance (theme, animation) to Settings?	Approve; it is required by your decisions.
Font	Cairo was unavailable here; mock-ups use Inter and DejaVu Sans stand-ins.	Ship Cairo; re-check wrapping in Arabic at 200% scale.
Copy	Arabic strings in mock-ups are illustrative.	Have a native reviewer approve final strings.
Spec items	DN-1 accounts model, DN-6 remote capabilities, DN-7 IMAGE_LOCAL are still open in the spec.	The UI assumes the recommended defaults (§13.1, §13.7, §11.3).